

Lab Guide: Endpoint Security & Windows Systems Defense

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Course: Cyber Defense

Estimated Duration: 8–10 hours

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Lab Overview

This lab combines the infrastructure of the "Defense in Depth" topology with the endpoint security exercises from the "Windows Systems" module.

Note on Infrastructure: Re-creating the entire infrastructure (Phase 1) is OPTIONAL if you already have a functional Active Directory lab (Domain Controller + Client + Firewall). However, the steps below outline exactly how to build the required environment to ensure all exercises work correctly.

Phase 1: Infrastructure Setup (The Foundation)

Architecture Overview

- **Hypervisor:** VirtualBox
 - **Firewall:** pfSense (Gateway for all segments)
 - **LAN Zone (192.168.1.x):** Windows Server 2019 (Domain Controller) & Windows 10/11 (Client)
 - **WAN Zone:** Bridged to your physical Internet
-

Step 1: Virtual Network Configuration (VirtualBox)

Instructions:

1. Open VirtualBox
2. Go to **File > Host Network Manager**
3. Ensure **Bridged Adapter** is available (this uses your physical internet connection)
4. You do not need to create Host-Only networks here; we will define Internal Networks directly in the VM settings:
 - **lan:** For Windows Server and Client
 - **dmz:** For optional services (if used)

Step 2: pfSense Firewall Setup (The Router)

2.1 Create VM

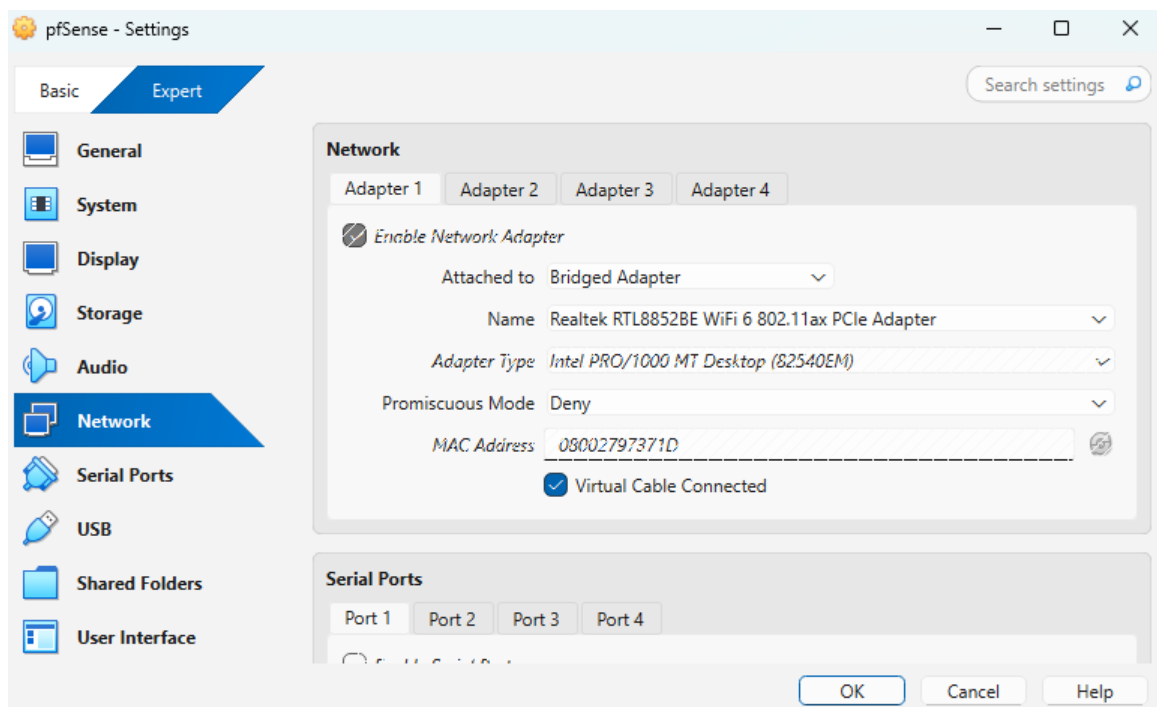
Settings:

- **Name:** pfSense
- **Type:** BSD
- **Version:** FreeBSD (64-bit)
- **RAM:** 1024 MB
- **CPU:** 1 core

2.2 Network Adapters Configuration

Adapter	Attached to	Network Name	Purpose
Adapter 1	Bridged Adapter	(Your physical NIC)	WAN (Internet)
Adapter 2	Internal Network	lan	LAN Zone
Adapter 3	Internal Network	dmz	DMZ/OPT1 Zone

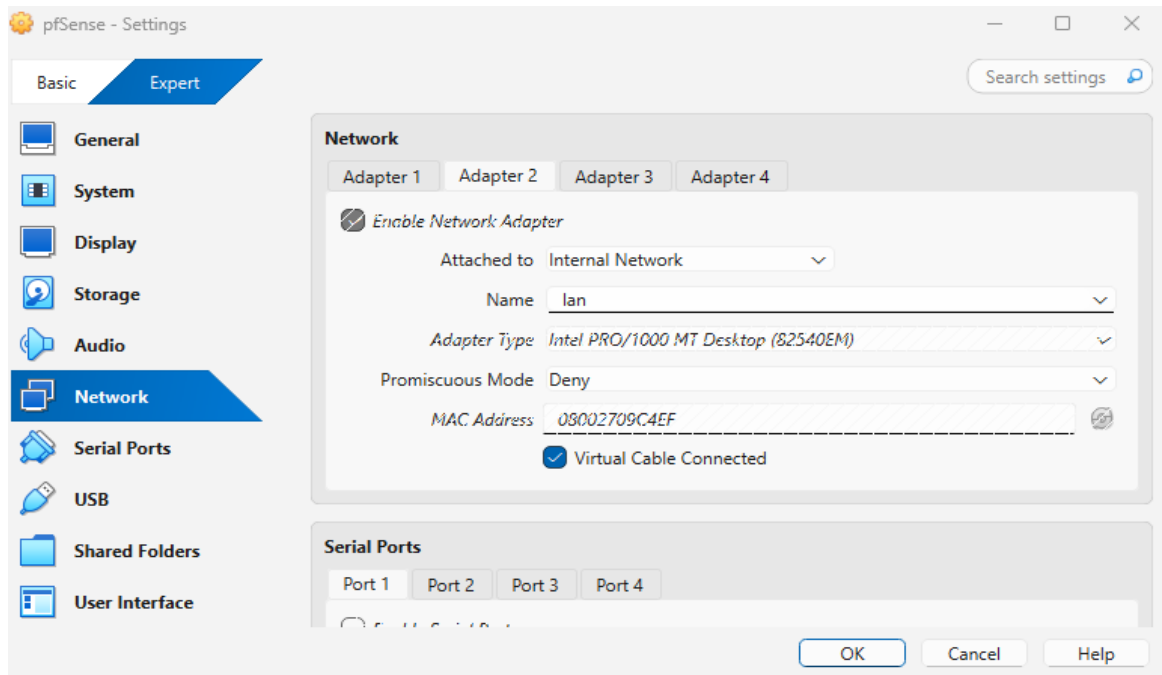
Table 1: Network Adapter Configuration for pfSense



01-pfsense-adapter1.png

Location: pfSense → Settings → Network → Adapter 1

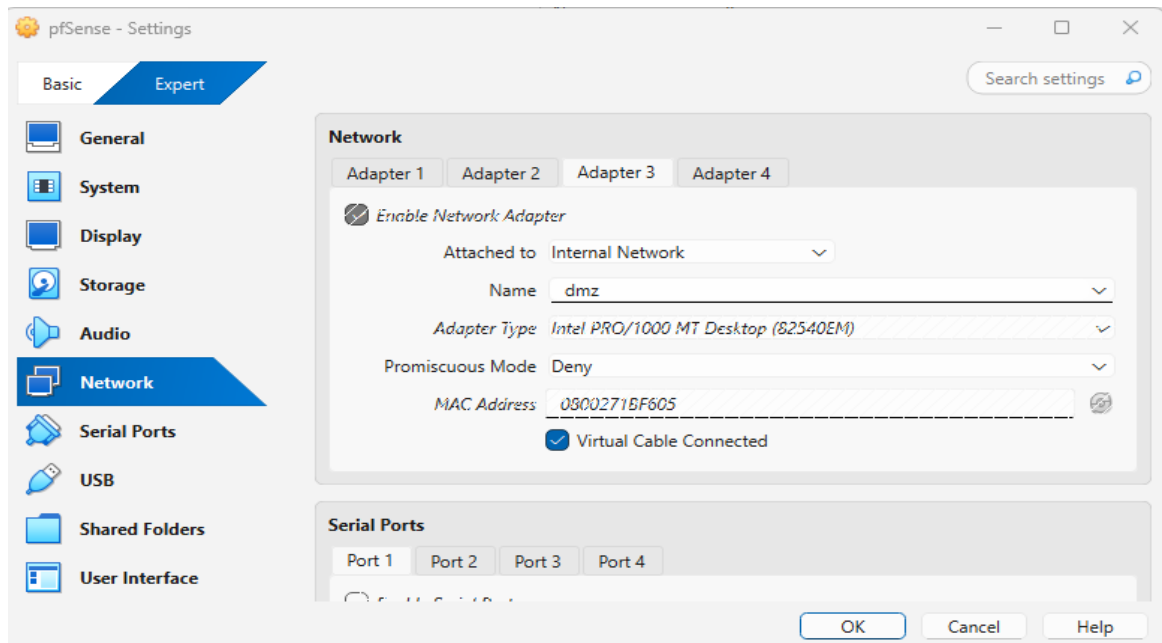
Show: Attached to: Bridged Adapter



02-pfsense-adapter2-lan.png

Location: pfSense → Settings → Network → Adapter 2

Show: Attached to: Internal Network, Name: lan



03-pfsense-adapter3-dmz.png

Location: pfSense → Settings → Network → Adapter 3
Show: Attached to: Internal Network, Name: dmz

2.3 Install pfSense

1. Mount the pfSense ISO
 2. Start the VM
 3. Follow the installation prompts:
 - Select **Install pfSense**
 - Choose **Default Keyboard**
 - Select **Auto (ZFS)** or **UFS**
 - Complete installation and **Reboot**
-

2.4 Configure IPs (Console Menu)

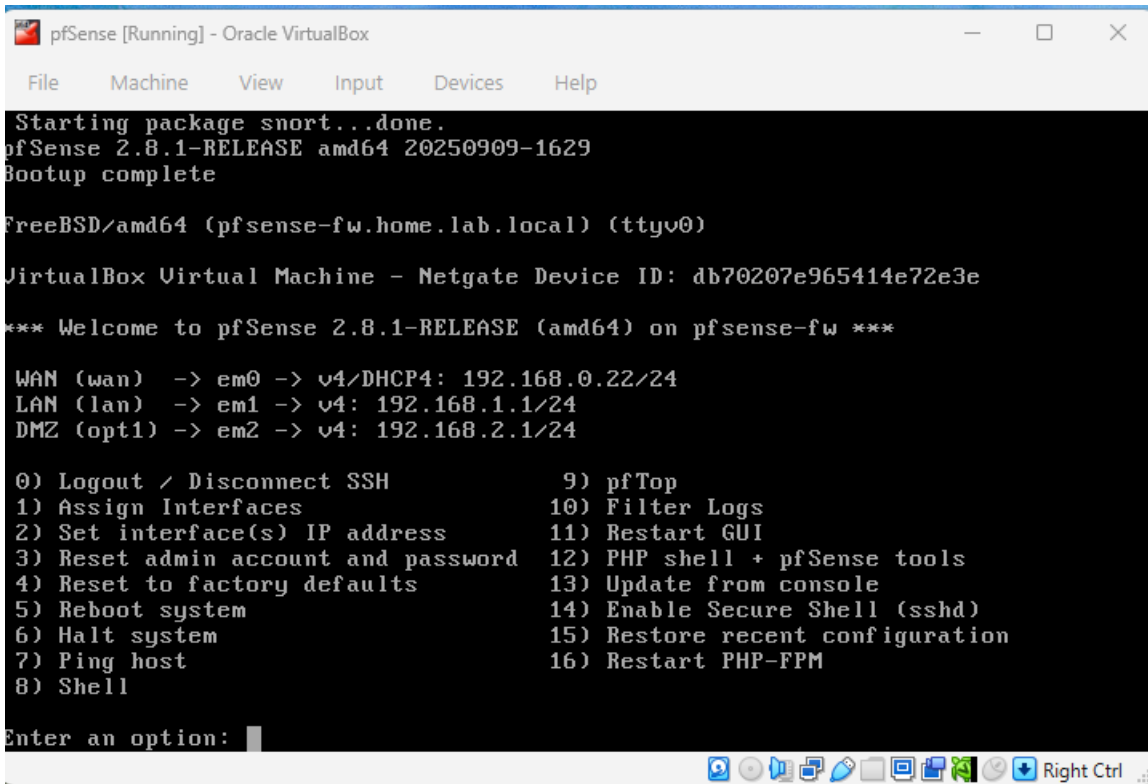
After reboot, configure interfaces via console:

WAN Configuration:

- DHCP: **Yes** (Auto-assigns IP from your home router)

LAN Configuration:

- Static IPv4: **192.168.1.1/24**
- Enable DHCP Server: **Yes**
- DHCP Range: **192.168.1.100 to 192.168.1.200**



```
pfSense [Running] - Oracle VirtualBox
File Machine View Input Devices Help

Starting package snort...done.
pfSense 2.8.1-RELEASE amd64 20250909-1629
Bootup complete

FreeBSD/amd64 (pfsense-fw.home.lab.local) (ttyv0)
VirtualBox Virtual Machine - Netgate Device ID: db70207e965414e72e3e

*** Welcome to pfSense 2.8.1-RELEASE (amd64) on pfsense-fw ***

WAN (wan) -> em0 -> v4/DHCP4: 192.168.0.22/24
LAN (lan) -> em1 -> v4: 192.168.1.1/24
DMZ (opt1) -> em2 -> v4: 192.168.2.1/24

0) Logout / Disconnect SSH
1) Assign Interfaces
2) Set interface(s) IP address
3) Reset admin account and password
4) Reset to factory defaults
5) Reboot system
6) Halt system
7) Ping host
8) Shell
9) pfTop
10) Filter Logs
11) Restart GUI
12) PHP shell + pfSense tools
13) Update from console
14) Enable Secure Shell (sshd)
15) Restore recent configuration
16) Restart PHP-FPM

Enter an option: 
```

04-pfsense-ip-config.png

Location: pfSense Console

Show: IP assignments for WAN (DHCP) and LAN (192.168.1.1/24)

2.5 WebGUI Setup

1. From a VM on the LAN (once created), access: <https://192.168.1.1>
2. Default credentials:
 - Username: `admin`
 - Password: `pfsense`

Critical Configuration:

- Go to **System** → **Advanced** → **Networking**
- **UNCHECK** "Block private networks" on WAN (Critical for home labs)

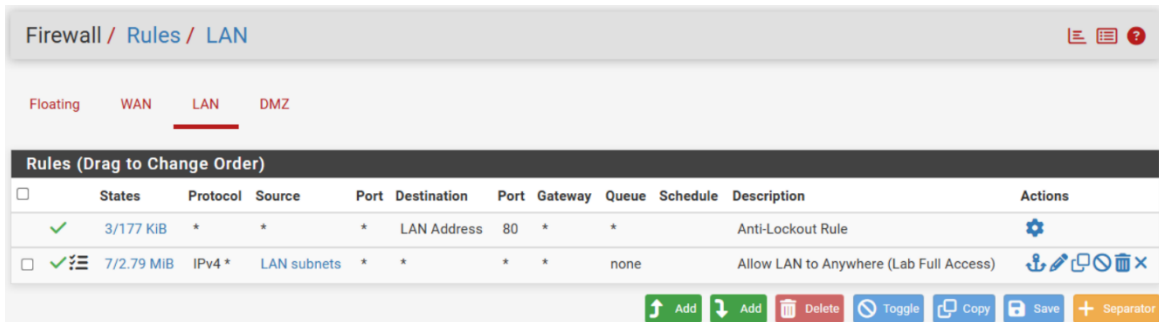


05-pfsense-unblock-private.png

Location: System → Advanced → Networking
Show: "Block private networks" UNCHECKED

Firewall Rules:

- Go to **Firewall** → **Rules** → **LAN**
- Ensure there is an **"Allow All"** rule (IPv4 * * * * *)



06-pfsense-lan-allow-all.png

Location: Firewall → Rules → LAN
Show: Allow IPv4 any → any rule

Step 3: Windows Server 2019 (The Domain Controller)

Refers to "AD Domain Controller" in exercises.

3.1 Create VM

Settings:

- **Name:** DC-Server
 - **OS:** Windows Server 2019
 - **RAM:** 4 GB
 - **Network:** Attached to: Internal Network, Name: lan
-

3.2 IP Configuration

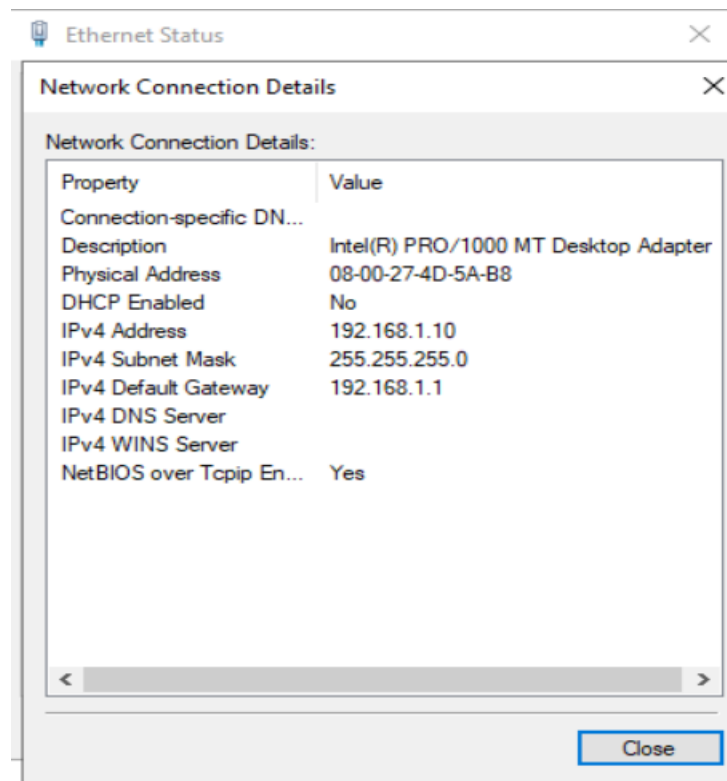
Set Static IP on Windows Server:

Setting	Value
IPv4 Address	192.168.1.10
Subnet Mask	255.255.255.0
Default Gateway	192.168.1.1
Preferred DNS	127.0.0.1 (Self)

Table 2: Windows Server Static IP Configuration

Steps:

1. Open **Control Panel** → **Network and Sharing Center**
2. Change adapter settings
3. Right-click **Ethernet** → **Properties** → **IPv4** → **Properties**
4. Enter static IP details above



07-dc-ip-config.png

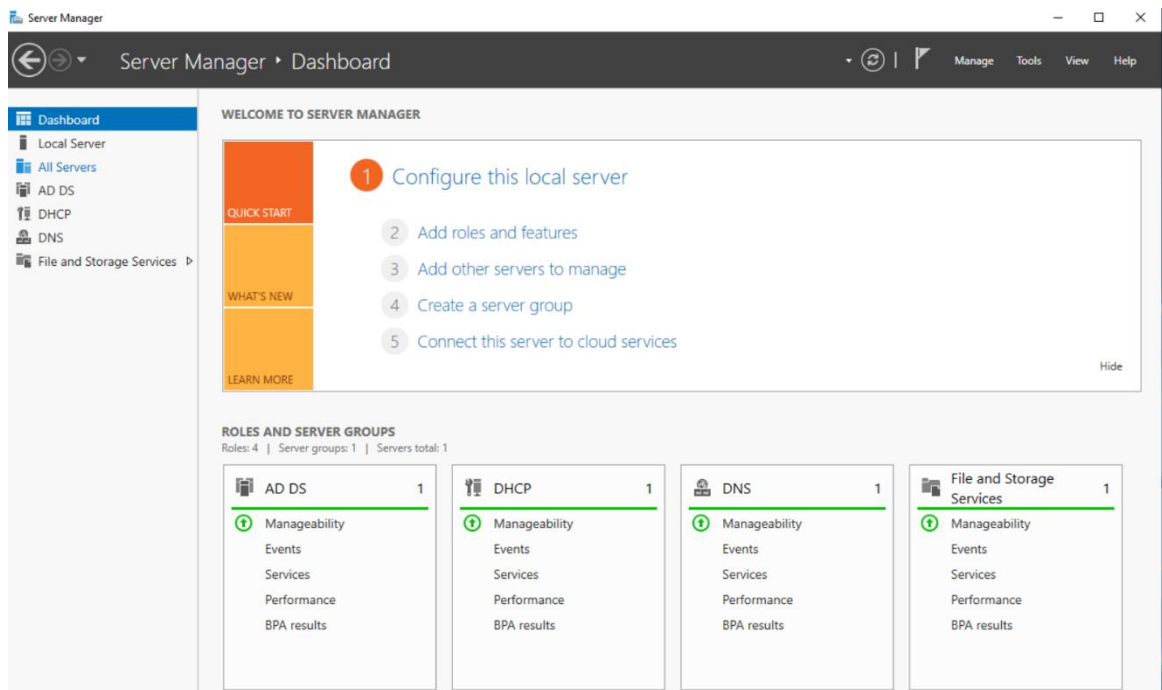
Location: Control Panel → Network → Ethernet → Status → Details
Show: IPv4: 192.168.1.10, Gateway: 192.168.1.1

3.3 Install Active Directory Domain Services

1. Open **Server Manager** → **Add Roles and Features**
 2. Select **Active Directory Domain Services**
 3. Click **Install**
-

3.4 Promote to Domain Controller

1. Click the **flag icon** in Server Manager
2. Select **Promote this server to a Domain Controller**
3. Choose **Add a new forest**
4. Domain name: **cnd.local**
5. DSRM Password: **Pa\$\$w0rd**
6. Complete the wizard and restart



08-ad-installed.png

Location: Server Manager Dashboard
Show: AD DS role installed, Domain: cnd.local

Step 4: Windows 10/11 Client (The User Machine)

Refers to "Admin Machine-1", "Finance-Dept", or "Web Server" (Client role) in exercises.

4.1 Create VM

Settings:

- **Name:** Win-Client
 - **OS:** Windows 10 or 11
 - **RAM:** 4 GB
 - **Network:** Attached to: Internal Network, Name: lan
-

4.2 Domain Join

Set DNS Server:

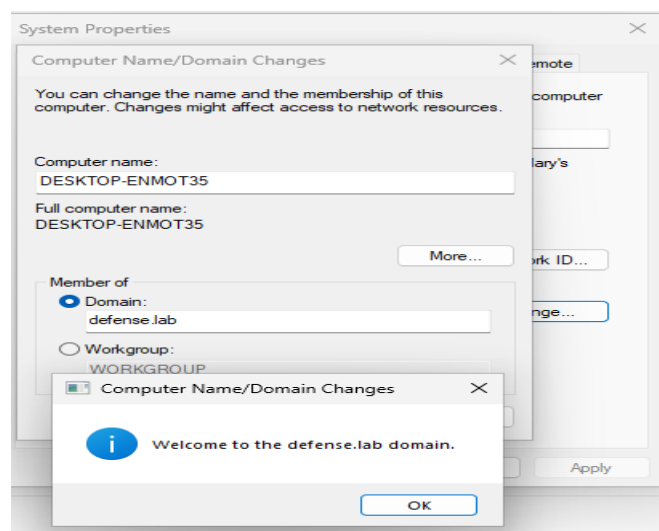
1. Set DNS server to **192.168.1.10** (The DC)

Join Domain:

1. **System Properties** → **Change Settings** → **Member of Domain:** cnd.local
2. Credentials:
 - Username: Administrator
 - Password: Pa\$\$w0rd

Location: CMD → ipconfig /all

Show: IPv4 (192.168.1.100-200), Gateway: 192.168.1.1, DNS: 192.168.1.10



09-client-domain-joined.png

Location: System → About
Show: Domain: cnd.local

4.3 Create Local User

Create a local user named **Admin** for troubleshooting tasks:

- Username: Admin
 - Password: admin@123
-

IP Addressing Summary Table

Machine	Adapter	Network	IP Address
pfSense	WAN	Bridged	DHCP (from ISP)
pfSense	LAN	Internal (lan)	192.168.1.1
pfSense	DMZ	Internal (dmz)	Not Used
DC Server	Ethernet	LAN	192.168.1.10
Client	Ethernet	LAN	DHCP (192.168.1.100+)

Table 3: IP Addressing Summary

Phase 2: Endpoint Security Exercises

Exercise 1: Basic Network Administration & Troubleshooting

Target: Windows Client ("Admin Machine-1")

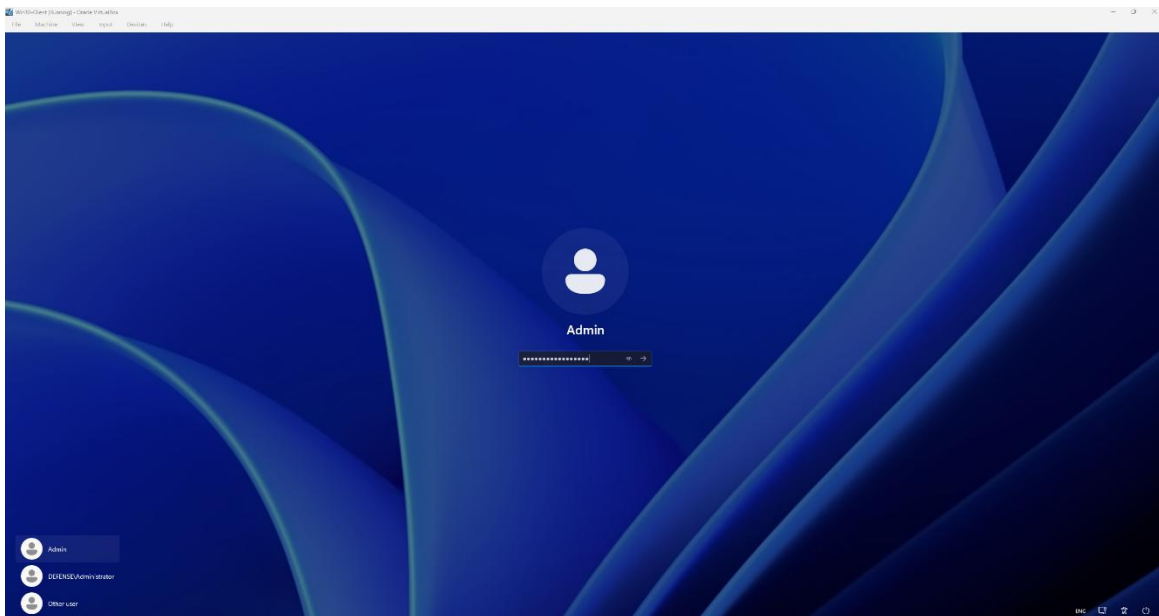
Objective:

- Practice basic network management
 - Diagnose connectivity issues
 - Execute fundamental network commands
-

Step 1: Login to Windows Client

Credentials:

- Username: Admin
- Password: admin@123



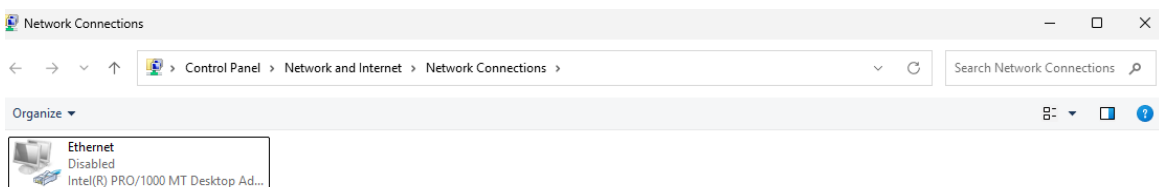
E1-01-client-login.png

Location: Desktop after login
Show: Logged in as local Admin user

Step 2: Disable Network Adapter

Steps:

1. Open **Control Panel** → **Network and Sharing Center**
2. Click **Change adapter settings**
3. Right-click **Ethernet** → **Disable**



E1-02-adapter-disabled.png

Location: Network Connections
Show: Ethernet adapter grayed out (disabled)

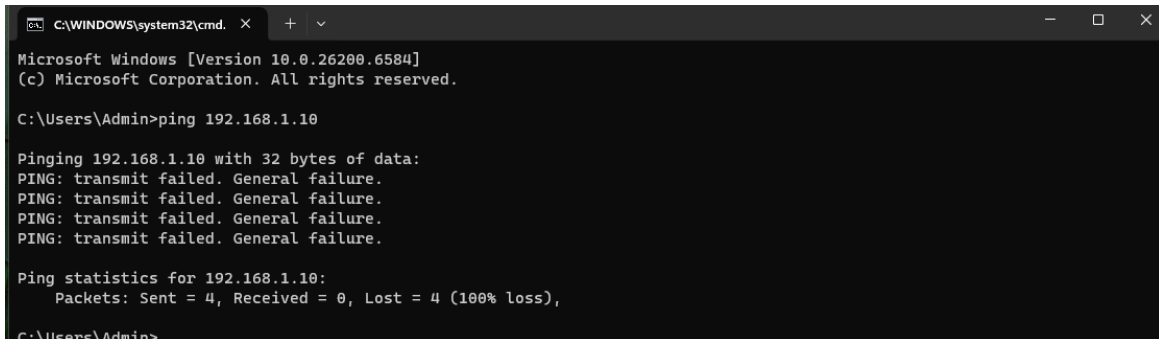
Step 3: Test Connectivity (Adapter Disabled)

Command:

ping 192.168.1.10

Expected Result:

- "Destination host unreachable" or "General failure"



```
C:\WINDOWS\system32\cmd. X + -
Microsoft Windows [Version 10.0.26200.6584]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Admin>ping 192.168.1.10

Pinging 192.168.1.10 with 32 bytes of data:
PING: transmit failed. General failure.
PING: transmit failed. General failure.
PING: transmit failed. General failure.
PING: transmit failed. General failure.

Ping statistics for 192.168.1.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\Users\Admin>
```

E1-03-ping-failed.png

Location: CMD

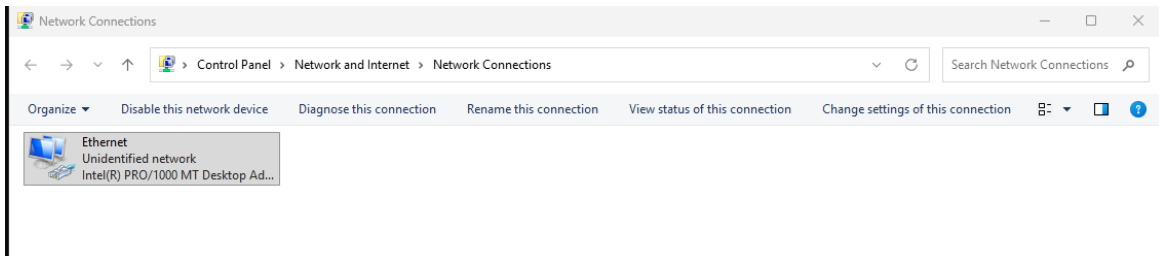
Show: Failed ping output

🛡️ **Security Note:** Disabling the Network Adapter breaks Layer 2 and Layer 3 connectivity completely.

Step 4: Enable Network Adapter

Steps:

1. Right-click **Ethernet** → **Enable**



E1-04-adapter-enabled.png

Location: Network Connections

Show: Ethernet adapter enabled

Step 5: Run Network Commands

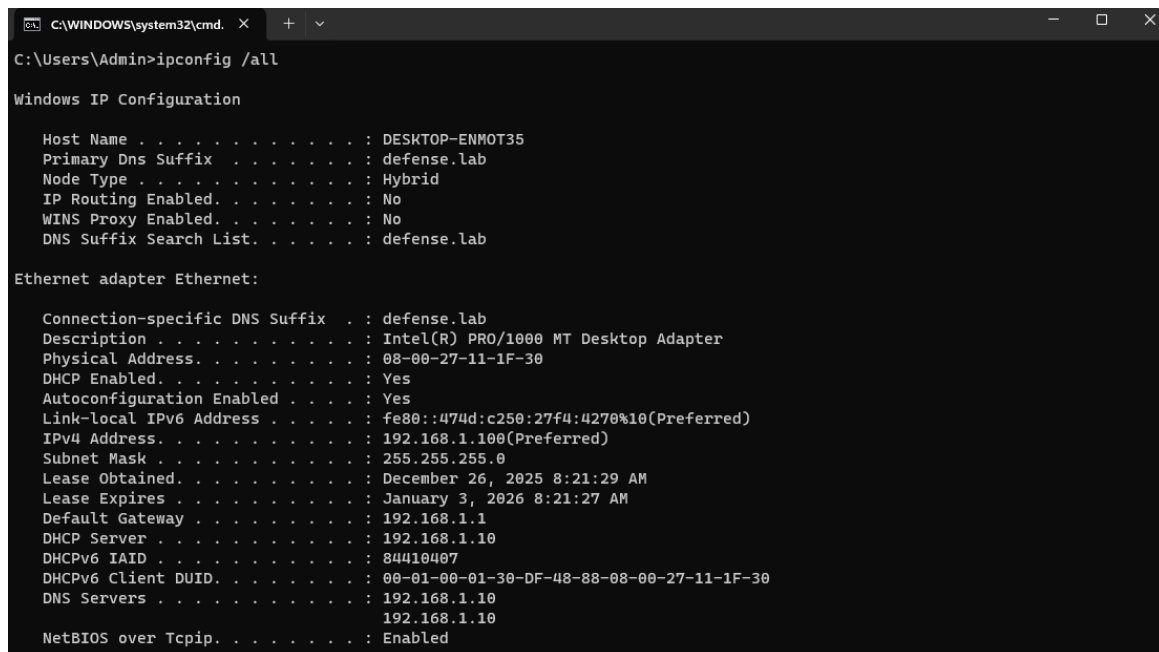
5.1 ipconfig /all

ipconfig /all

Expected Output:

Field	Expected Value
IPv4 Address	192.168.1.100
Subnet Mask	255.255.255.0
Default Gateway	192.168.1.1
DNS Server	192.168.1.10
DHCP Enabled	Yes

Table 4: ipconfig /all Expected Output



```
C:\WINDOWS\system32\cmd. x
C:\Users\Admin>ipconfig /all

Windows IP Configuration

Host Name . . . . . : DESKTOP-ENMOT35
Primary Dns Suffix . . . . . : defense.lab
Node Type . . . . . : Hybrid
IP Routing Enabled. . . . . : No
WINS Proxy Enabled. . . . . : No
DNS Suffix Search List. . . . . : defense.lab

Ethernet adapter Ethernet:

Connection-specific DNS Suffix . : defense.lab
Description . . . . . : Intel(R) PRO/1000 MT Desktop Adapter
Physical Address. . . . . : 08-00-27-11-1F-30
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
Link-local IPv6 Address . . . . . : fe80::474d:c250:27f4:4270%10(Preferred)
IPv4 Address. . . . . : 192.168.1.100(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Lease Obtained. . . . . : December 26, 2025 8:21:29 AM
Lease Expires . . . . . : January 3, 2026 8:21:27 AM
Default Gateway . . . . . : 192.168.1.1
DHCP Server . . . . . : 192.168.1.10
DHCPv6 IAID . . . . . : 84410407
DHCPv6 Client DUID. . . . . : 00-01-00-01-30-DF-48-88-08-00-27-11-1F-30
DNS Servers . . . . . : 192.168.1.10
                        192.168.1.10
NetBIOS over Tcpip. . . . . : Enabled
```

E1-05-ipconfig.png

Location: CMD

Show: Full output of ipconfig /all

5.2 ping Domain Controller

ping 192.168.1.10

Expected Result: Reply from 192.168.1.10

```
C:\Users\Admin>ping 192.168.1.10

Pinging 192.168.1.10 with 32 bytes of data:
Reply from 192.168.1.10: bytes=32 time=14ms TTL=128
Reply from 192.168.1.10: bytes=32 time=16ms TTL=128
Reply from 192.168.1.10: bytes=32 time=6ms TTL=128
Reply from 192.168.1.10: bytes=32 time=50ms TTL=128

Ping statistics for 192.168.1.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 6ms, Maximum = 50ms, Average = 21ms
```

E1-06-ping-dc.png

Location: CMD

Show: Successful ping to DC

5.3 tracert Internet

tracert 8.8.8.8

Expected Result: First hop should be 192.168.1.1 (pfSense)

```
C:\Users\Admin>tracert 8.8.8.8

Tracing route to dns.google [8.8.8.8]
over a maximum of 30 hops:

  0  0 ms  0 ms  0 ms  192.168.1.1
  1  8 ms  6 ms  7 ms  192.168.0.1
  2 18 ms 21 ms 22 ms modemcable001.133-130-66.mc.videotron.ca [66.130.133.1]
  3 18 ms 21 ms 22 ms 10.170.193.82
  4 20 ms 33 ms 28 ms 72.14.205.46
  5 19 ms 34 ms 30 ms 192.178.86.253
  6 20 ms 21 ms 28 ms 142.250.58.131
  7 28 ms 18 ms 18 ms dns.google [8.8.8.8]

Trace complete.
```

E1-07-tracert.png

Location: CMD

Show: Traceroute showing pfSense as first hop

🔒 **Security Note:** This confirms all traffic passes through the firewall.

5.4 netstat -an

netstat -an

Shows: Active connections and listening ports

```
C:\Users\Admin>netstat -an

Active Connections

Proto Local Address           Foreign Address         State
TCP 0.0.0.0:115             0.0.0.0*                LISTENING
TCP 0.0.0.0:445             0.0.0.0*                LISTENING
TCP 0.0.0.0:5444          0.0.0.0*                LISTENING
TCP 0.0.0.0:7480          0.0.0.0*                LISTENING
TCP 0.0.0.0:48444         0.0.0.0*                LISTENING
TCP 0.0.0.0:48445         0.0.0.0*                LISTENING
TCP 0.0.0.0:48446         0.0.0.0*                LISTENING
TCP 0.0.0.0:48470         0.0.0.0*                LISTENING
TCP 0.0.0.0:48471         0.0.0.0*                LISTENING
TCP 0.0.0.0:48472         0.0.0.0*                LISTENING
TCP 0.0.0.0:48782         0.0.0.0*                LISTENING
TCP 192.168.1.100:239     0.0.0.0*                LISTENING
TCP 192.168.1.100:50487  66.150.42.97:443        CLOSE_WAIT
TCP 192.168.1.100:50488  23.222.17.203:443       CLOSE_WAIT
TCP 192.168.1.100:50489  23.222.17.203:443       CLOSE_WAIT
TCP 192.168.1.100:50490  23.222.17.203:443       CLOSE_WAIT
TCP 192.168.1.100:50491  23.222.17.207:443       CLOSE_WAIT
TCP 192.168.1.100:50492  23.10.131.18:80         ESTABLISHED
TCP 192.168.1.100:50493  23.10.131.18:80         ESTABLISHED
TCP 192.168.1.100:50494  150.232.92.34:443       TIME_WAIT
TCP 192.168.1.100:50495  66.150.42.97:80         TIME_WAIT
TCP 192.168.1.100:50496  82.121.126.16:443       ESTABLISHED
TCP 192.168.1.100:50497  151.101.138.172:80      SYN_SENT
TCP 192.168.1.100:50498  151.101.138.172:80      SYN_SENT
TCP 192.168.1.100:50499  151.101.138.172:80      SYN_SENT
TCP 192.168.1.100:50500  151.101.138.172:80      SYN_SENT
TCP 192.168.1.100:50501  151.101.138.172:80      SYN_SENT
TCP [::]:115             [::]:0                  LISTENING
TCP [::]:445            [::]:0                  LISTENING
TCP [::]:7480           [::]:0                  LISTENING
TCP [::]:48444          [::]:0                  LISTENING
TCP [::]:48445          [::]:0                  LISTENING
TCP [::]:48446          [::]:0                  LISTENING
TCP [::]:48470          [::]:0                  LISTENING
TCP [::]:48471          [::]:0                  LISTENING
TCP [::]:48472          [::]:0                  LISTENING
TCP [::]:48782          [::]:0                  LISTENING
TCP [::]:48480          [::]:0                  LISTENING
TCP [::]:42080          [::]:0                  LISTENING
UDP 0.0.0.0:112          0.0.0.0*                *.*
UDP 0.0.0.0:5800        0.0.0.0*                *.*
UDP 0.0.0.0:5101        0.0.0.0*                *.*
UDP 0.0.0.0:5103        0.0.0.0*                *.*
UDP 0.0.0.0:56000       0.0.0.0*                *.*
UDP 0.0.0.0:51376      0.0.0.0*                *.*
UDP 127.0.0.1:1000      127.0.0.1*              *.*
UDP 127.0.0.1:53038    127.0.0.1:53038         *.*
UDP 127.0.0.1:53039    127.0.0.1:53039         *.*
UDP 127.0.0.1:56057    127.0.0.1:56057         *.*
UDP 127.0.0.1:58000    127.0.0.1:58000         *.*
UDP 127.0.0.1:51081    127.0.0.1:51081         *.*
UDP 127.0.0.1:51083    127.0.0.1:51083         *.*
UDP 127.0.0.1:51051    127.0.0.1:51051         *.*
UDP 127.0.0.1:68888    0.0.0.0*                *.*
```

E1-08-netstat.png

Location: CMD
Show: List of listening ports and established connections

5.5 arp -a

arp -a

Expected Result: Should show MAC address mapping for 192.168.1.1

```
C:\Users\Admin>arp -a

Interface: 192.168.1.100 --- 0xa
Internet Address      Physical Address      Type
192.168.1.1           08-00-27-09-c4-ef     dynamic
192.168.1.10          08-00-27-4d-5a-b8     dynamic
192.168.1.255          ff-ff-ff-ff-ff-ff     static
224.0.0.22            01-00-5e-00-00-16     static
224.0.0.251           01-00-5e-00-00-fb     static
224.0.0.252           01-00-5e-00-00-fc     static
239.255.255.250       01-00-5e-7f-ff-fa     static
255.255.255.255       ff-ff-ff-ff-ff-ff     static
```

E1-09-arp.png

Location: CMD
Show: ARP cache including gateway MAC

Command Summary Table

Command	Purpose
ipconfig /all	Display full IP configuration
ping	Test connectivity
tracert	Trace packet route
netstat -an	Show active connections and ports
arp -a	Display IP-to-MAC address mappings

Table 5: Network Troubleshooting Commands

Exercise 2: Securing Windows File Share in Active Directory

Targets: Windows Server 2019 & Windows Client

Objective:

- Implement Role-Based Access Control (RBAC)
- Use NTFS Permissions + AD Security Groups
- Enable Access-Based Enumeration (ABE)

Step 1: Create Security Group & Users (Server)

Location: Windows Server 2019

Steps:

1. Open **Server Manager** → **Tools** → **Active Directory Users and Computers**
2. Create two users:

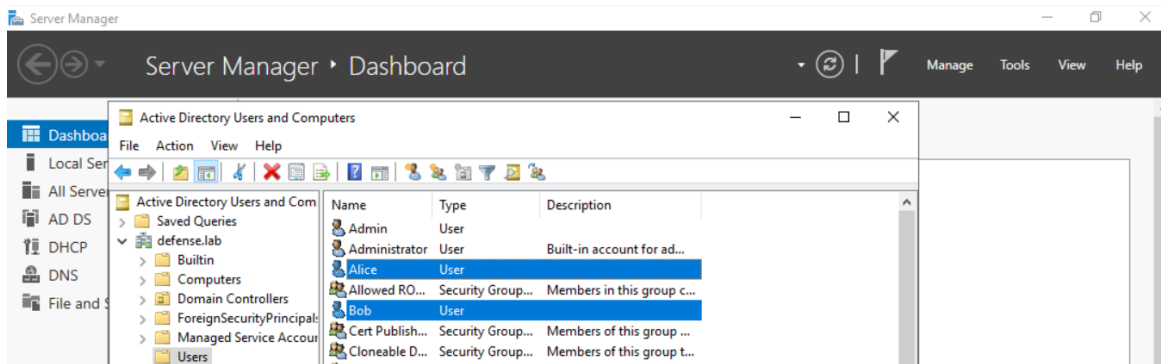
Username	Password
Bob	Pa\$\$w0rd
Alice	Pa\$\$w0rd

Table 6: AD Users to Create

3. Create Security Group:
 - Name: **FinanceData**
 - Group Scope: **Global**

- Group Type: **Security**

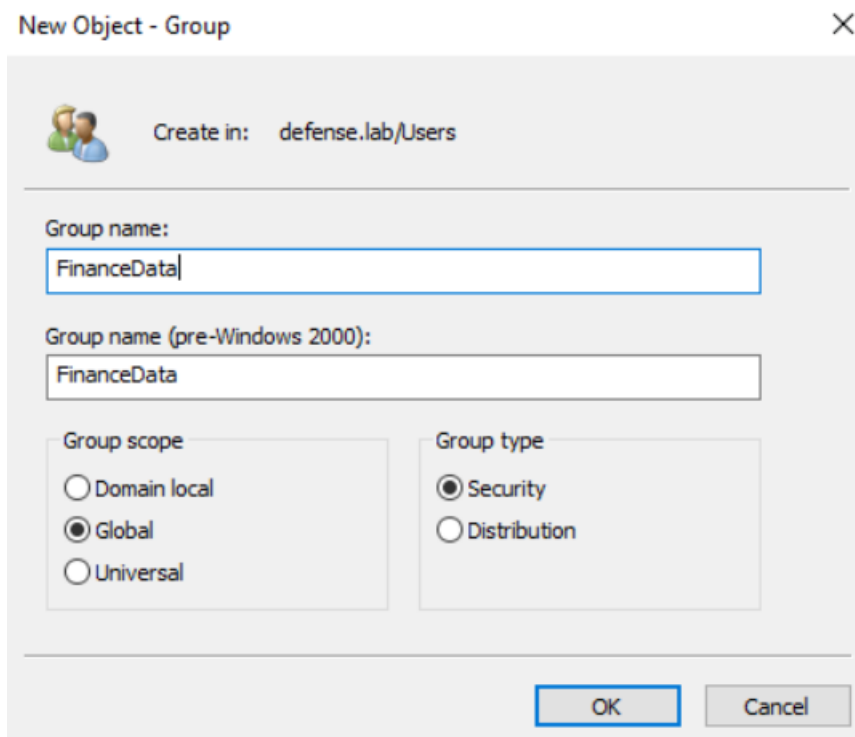
4. Add **Bob** to **FinanceData** group



E2-01-ad-users.png

Location: Active Directory Users and Computers

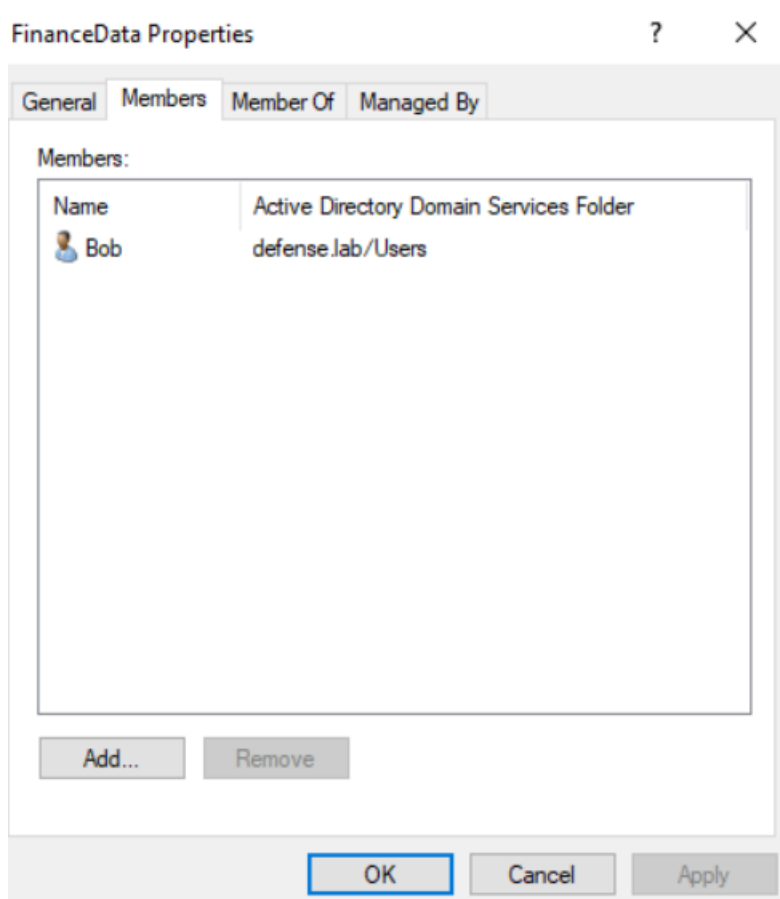
Show: Bob and Alice users created



E2-02-finance-group.png

Location: Active Directory Users and Computers

Show: FinanceData security group



E2-03-bob-in-group.png

Location: FinanceData Properties → Members

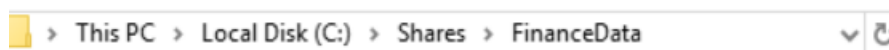
Show: Bob as member of FinanceData

🛡️ **Security Note:** Permissions are assigned to groups, not individual users (Best Practice).

Step 2: Create Share (Server)

Steps:

1. Create folder: C:\Shares\FinanceData

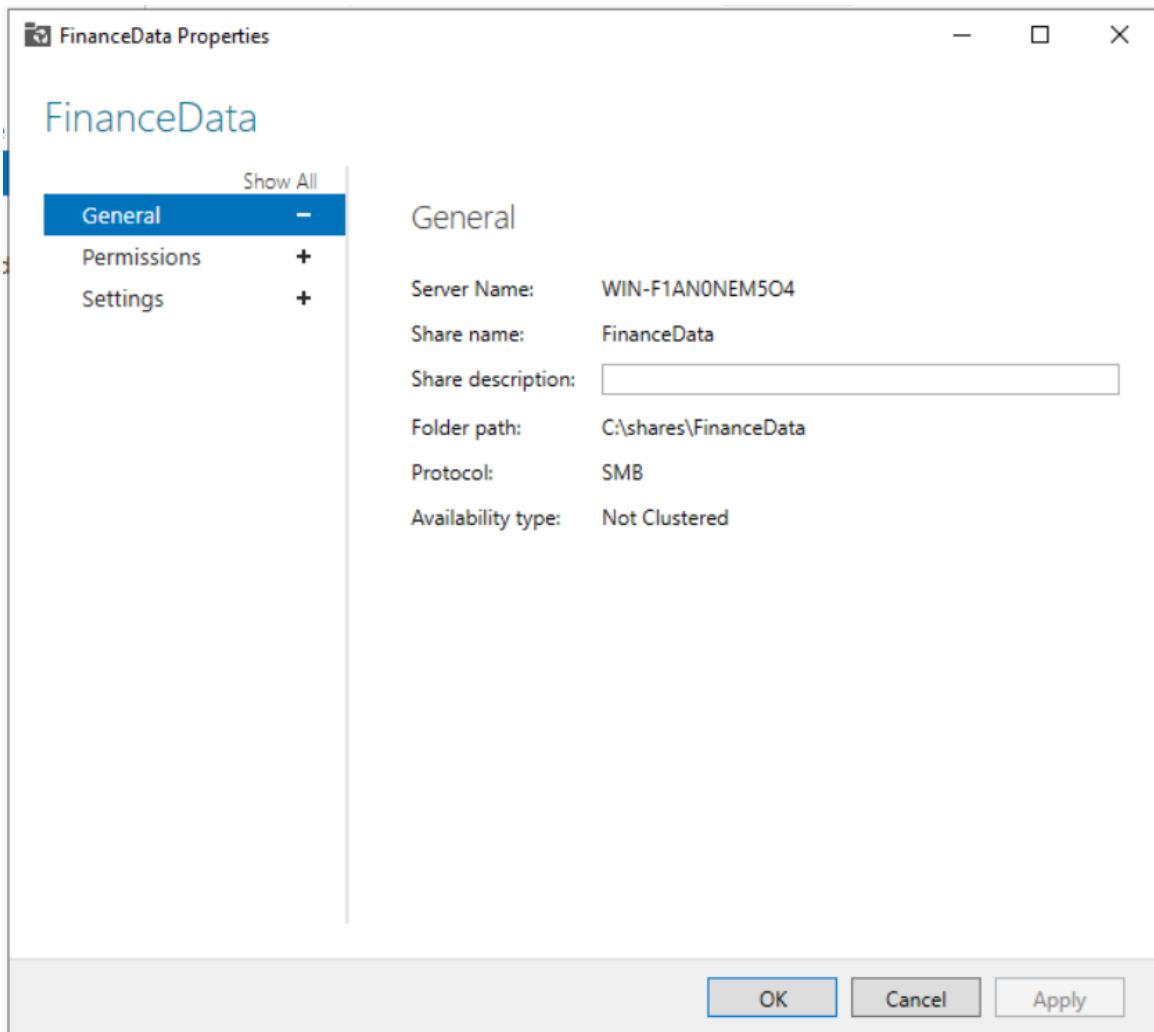


E2-04-finance-folder.png

Location: File Explorer

Show: C:\Shares\FinanceData folder created

2. Open **Server Manager** → **File and Storage Services** → **Shares**
3. Click **Tasks** → **New Share**
4. Select **SMB Share – Quick**
5. Configure:
 - Share Name: **FinanceData**
 - Share Location: C:\Shares\FinanceData
 - **✓ Enable access-based enumeration**



E2-05-share-abe.png

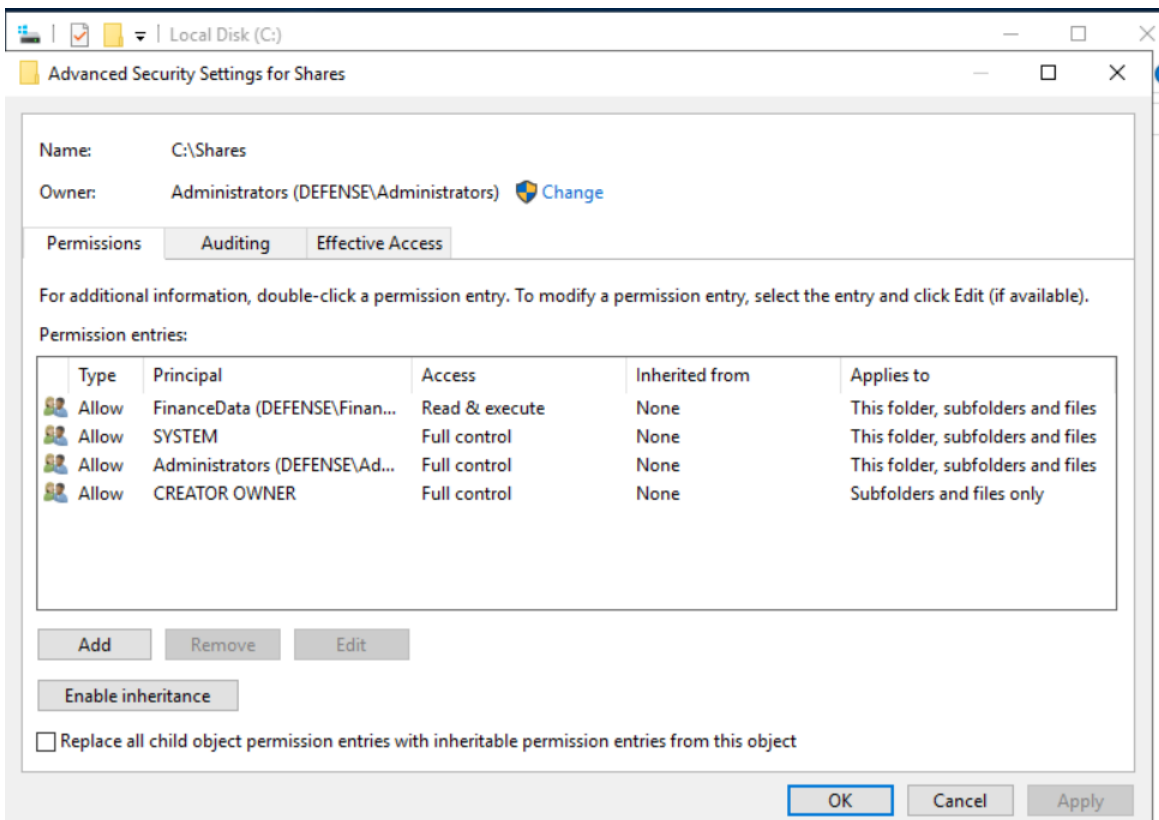
Location: New Share Wizard

Show: Access-based enumeration checkbox CHECKED

Step 3: Configure Permissions (Server)

Steps:

1. In the wizard, click **Customize permissions**
2. Click **Disable inheritance**
3. Choose **Convert inherited permissions into explicit permissions**
4. **Remove** the **Users** group
5. Click **Add** → **Select Principal** → **Type: FinanceData**
6. Grant **Modify** permissions



E2-06-ntfs-permissions.png

Location: Security → Advanced

Show: Only FinanceData group with Modify permissions

🔒 **Security Note:** NTFS + ABE combination ensures unauthorized users cannot even see the share.

Step 4: Verify Access (Client)

4.1 Test with Bob (Authorized)

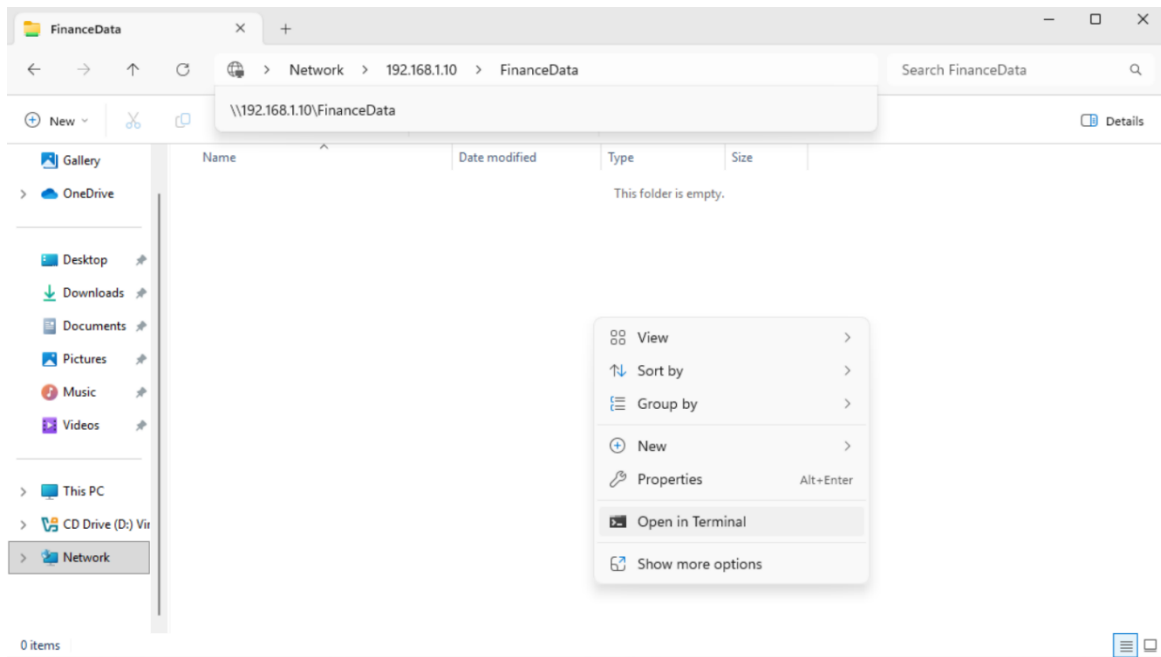
Login:

- Username: cnd\Bob
- Password: Pa\$\$w0rd

Access Share:

\\192.168.1.10\FinanceData

Expected Result: ✓ Access Granted



E2-07-bob-access.png

Location: File Explorer

Show: Bob successfully accessing FinanceData share

4.2 Test with Alice (Unauthorized)

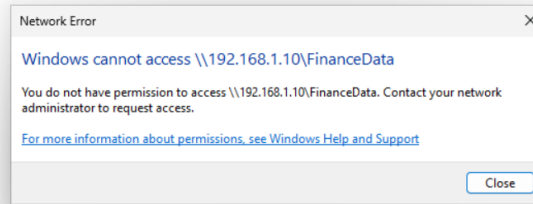
Login:

- Username: cnd\Alice
- Password: Pa\$\$w0rd

Access Share:

\\192.168.1.10\FinanceData

Expected Result: ✗ Access Denied (or folder not visible)



E2-08-alice-denied.png

Location: File Explorer or Access Denied dialog

Show: Alice cannot access or see the share

Access Control Summary Table

User	Group	Access
Bob	FinanceData	Allowed
Alice	None	Denied

Table 7: File Share Access Control Results

Exercise 3: Analyzing Security Baseline (MSCT)

Target: Windows Server 2019 (Domain Controller)

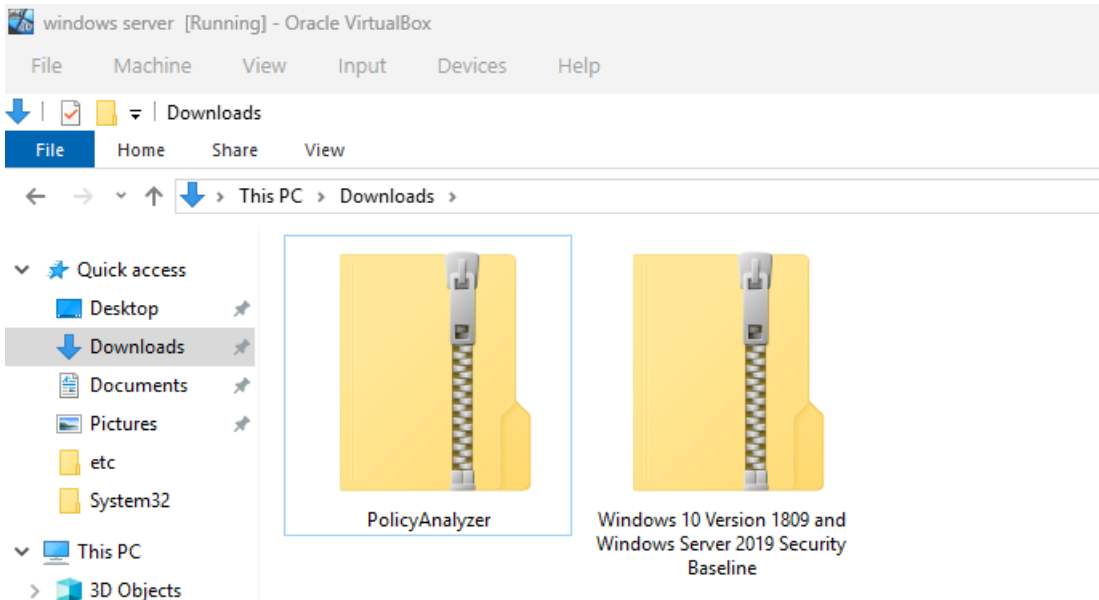
Objective:

- Assess current security posture
 - Compare against Microsoft Security Best Practices
 - Identify misconfigurations (Critical Blue Team skill)
-

Step 1: Download Tools

Files Required:

1. **PolicyAnalyzer.zip** - [Microsoft Security Compliance Toolkit](#)
2. **Windows Server 2019 Security Baseline.zip**



E3-01-download-files.png

Location: Downloads folder

Show: Both ZIP files downloaded

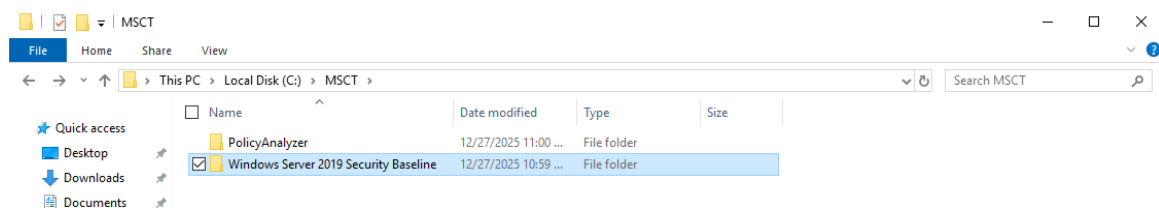
Step 2: Extract and Prepare

Recommended Path: C:\MSCT

Folder Structure:

C:\MSCT

├─ PolicyAnalyzer
└─ Windows Server 2019 Security Baseline\



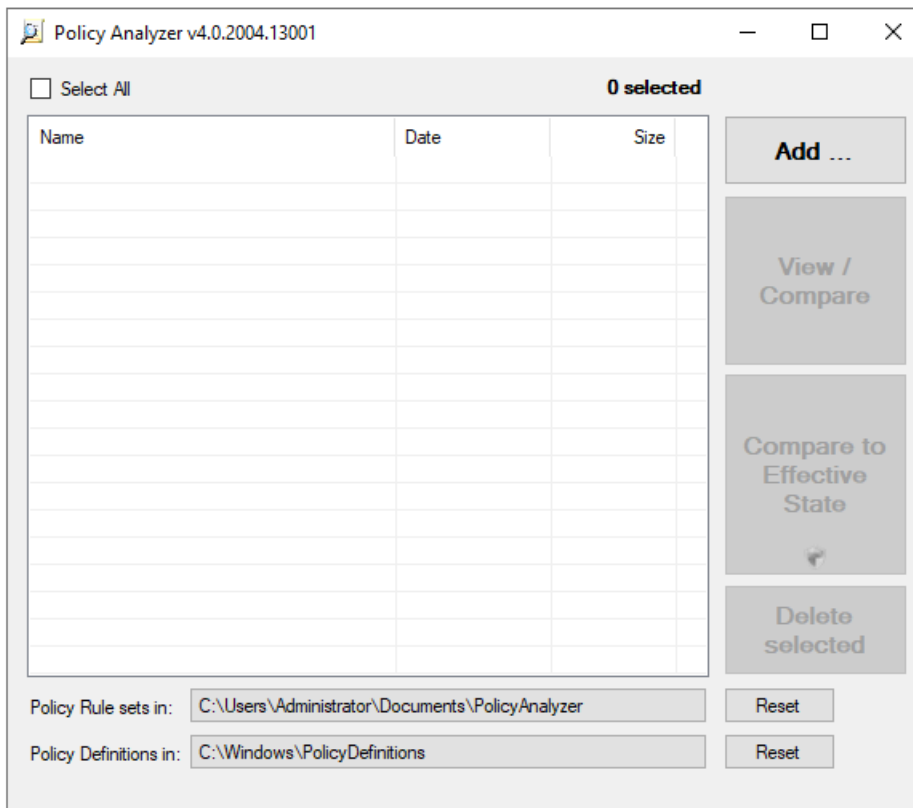
E3-02-extracted-folders.png

Location: File Explorer

Show: Folder structure after extraction

Step 3: Run Policy Analyzer

Execute: C:\MSCT\PolicyAnalyzer\PolicyAnalyzer.exe



E3-03-policy-analyzer-main.png

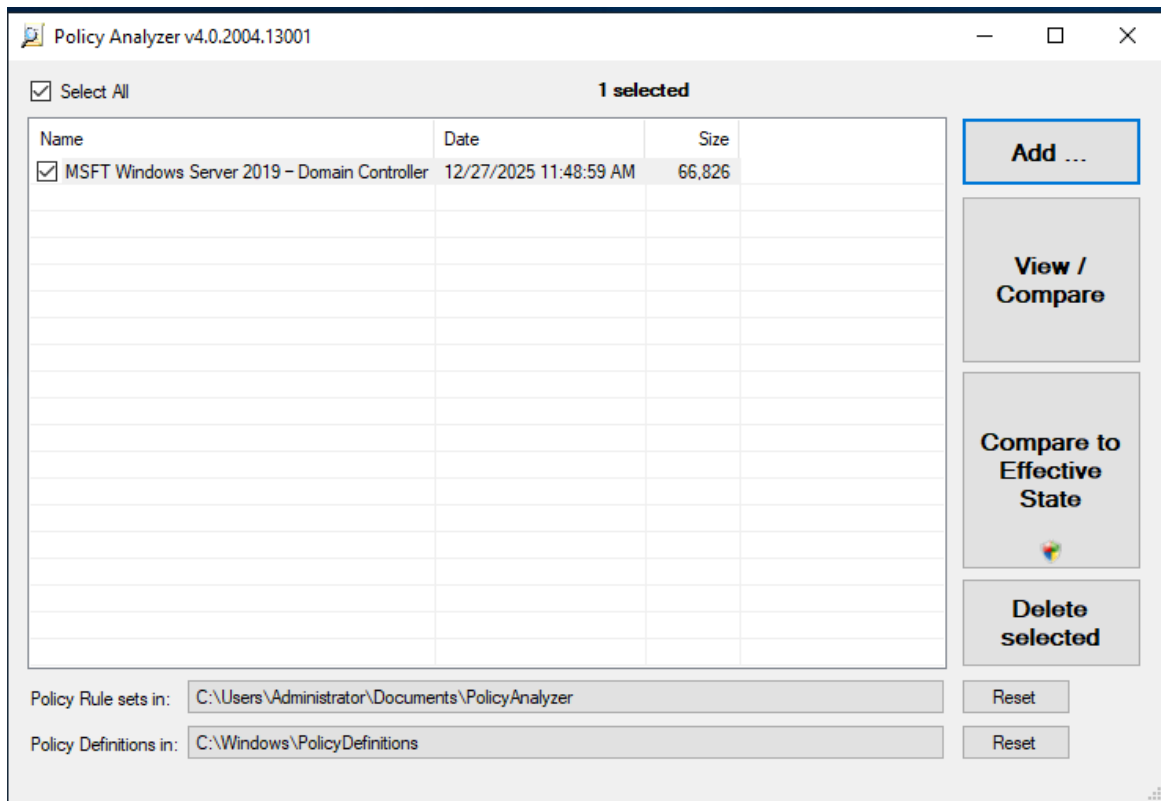
Location: PolicyAnalyzer window

Show: Main interface of Policy Analyzer

Step 4: Import Baseline

Steps:

1. Click **Add**
2. Browse to: Windows Server 2019 Security Baseline\GPOs
3. Click **Import**
4. Select: **MSFT Windows Server 2019 – Domain Controller**

**Location:** PolicyAnalyzer

Step 5: Compare with Local System

1. Select the imported Baseline
2. Click **Compare to Local Registry**

E3-05-compare-local.png

Step 6: Analyze Results

Color Coding:

Table 8: Policy Analyzer Color Coding

Policy Viewer - 156 items

Clipboard View Export Options

Policy Type	Policy Group or Registry Key	Policy Setting	MSFT Windows Server 2019 - Domain Controller
Security Template	Privilege Rights	SeEnableDelegationPrivilege	*S-1-5-32-544
Security Template	Privilege Rights	SeImpersonatePrivilege	*S-1-5-19,*S-1-5-...
Security Template	Privilege Rights	SeInteractiveLogonRight	*S-1-5-32-544
Security Template	Privilege Rights	SeLoadDriverPrivilege	*S-1-5-32-544
Security Template	Privilege Rights	SeLockMemoryPrivilege	
Security Template	Privilege Rights	SeManageVolumePrivilege	*S-1-5-32-544
Security Template	Privilege Rights	SeNetworkLogonRight	*S-1-5-11,*S-1-5-...
Security Template	Privilege Rights	SeProfileSingleProcessPrivilege	*S-1-5-32-544
Security Template	Privilege Rights	SeRemoteInteractiveLogonRight	*S-1-5-32-544
Security Template	Privilege Rights	SeRemoteShutdownPrivilege	*S-1-5-32-544
Security Template	Privilege Rights	SeRestorePrivilege	*S-1-5-32-544
Security Template	Privilege Rights	SeSecurityPrivilege	*S-1-5-32-544
Security Template	Privilege Rights	SeSystemEnvironmentPrivilege	*S-1-5-32-544
Security Template	Privilege Rights	SeTakeOwnershipPrivilege	*S-1-5-32-544
Security Template	Privilege Rights	SeTcbPrivilege	
Security Template	Privilege Rights	SeTrustedCredManAccessPrivilege	
Security Template	Service General Setting	"AppIDSvc"	2,""
Security Template	System Access	LSAAnonymousNameLookup	0

Policy Path:
 Advanced Audit Policy Configuration
 System Audit Policies\Account Logon
 Credential Validation

Credential Validation

This policy setting allows you to audit events generated by validation tests on user account logon credentials.

Events in this subcategory occur only on the computer that is authoritative for those credentials. For domain accounts, the domain controller is authoritative. For local accounts, the local computer is authoritative.

Volume: High on domain controllers.

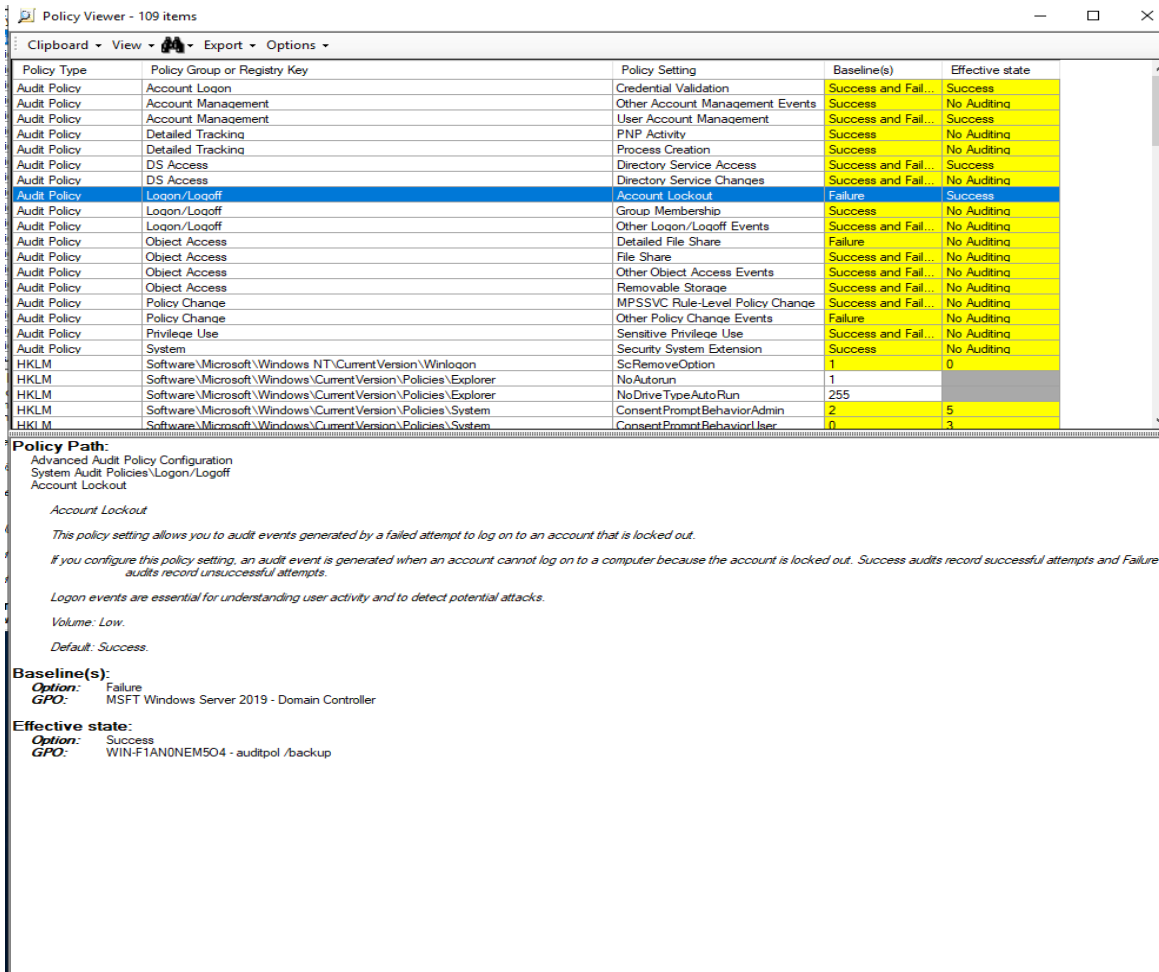
Default on Client editions: No Auditing.

Default on Server editions: Success.

MSFT Windows Server 2019 - Domain Controller:
Option: Success and Failure

E3-06-analysis-results.png

Location: PolicyAnalyzer Results
Show: List of discrepancies



E3-07-red-setting.png

Location: PolicyAnalyzer

Show: Example of a red (non-compliant) setting selected

Common Misconfigurations Found

Setting	Baseline	Local	Status
Minimum Password Length	14	8	Non-Compliant
SMB Signing	Enabled	Disabled	Non-Compliant
Audit Logon Events	Enabled	Not Configured	Warning

Table 9: Common Baseline Misconfigurations

🛡️ **Security Analysis:** These findings highlight areas where the Domain Controller deviates from Microsoft's security best practices and requires hardening.

Exercise 4: Remote Patch Management (BatchPatch)

Targets: Windows Client (Admin Console) & Windows Server 2019 (Target)

Objective:

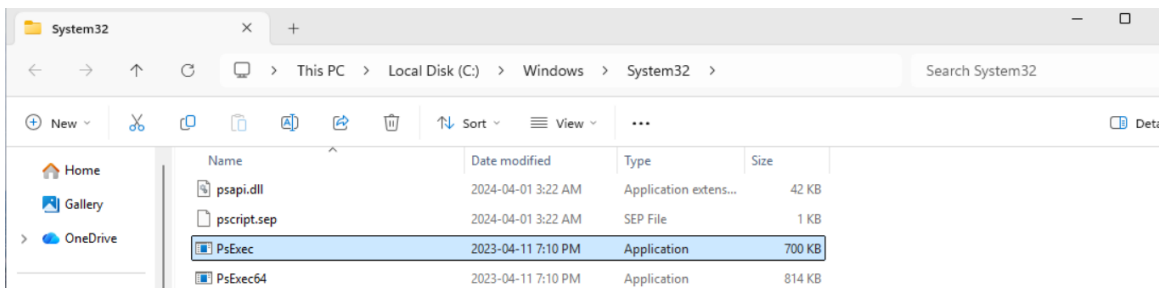
- Manage patches remotely
 - Identify missing security updates
 - Simulate real SysAdmin/Blue Team workflow
-

Step 1: Install PsExec (Prerequisite)

Location: Windows Client

Steps:

1. Download **PSTools** from [Sysinternals](#)
2. Extract the ZIP file
3. Copy PsExec.exe to: C:\Windows\System32\



E4-01-psexec-installed.png

Location: C:\Windows\System32

Show: PsExec.exe in System32 folder

🛡️ **Security Note:** BatchPatch uses PsExec for remote command execution.

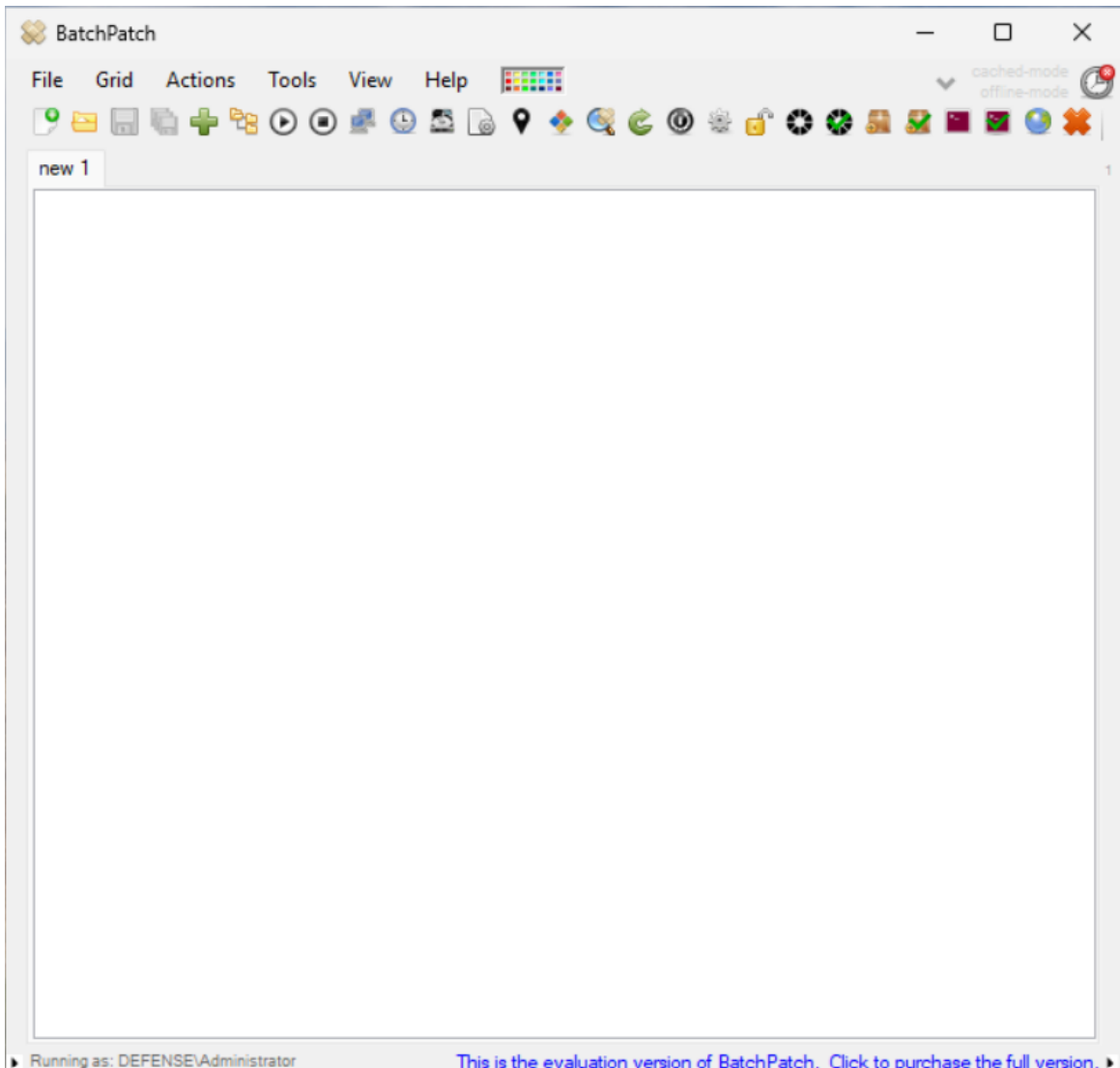
Step 2: Download and Run BatchPatch

Location: Windows Client

Download: [BatchPatch](#) (Free Evaluation)

Steps:

1. Download and install BatchPatch
2. Run as Administrator



E4-02-batchpatch-main.png

Location: BatchPatch main window

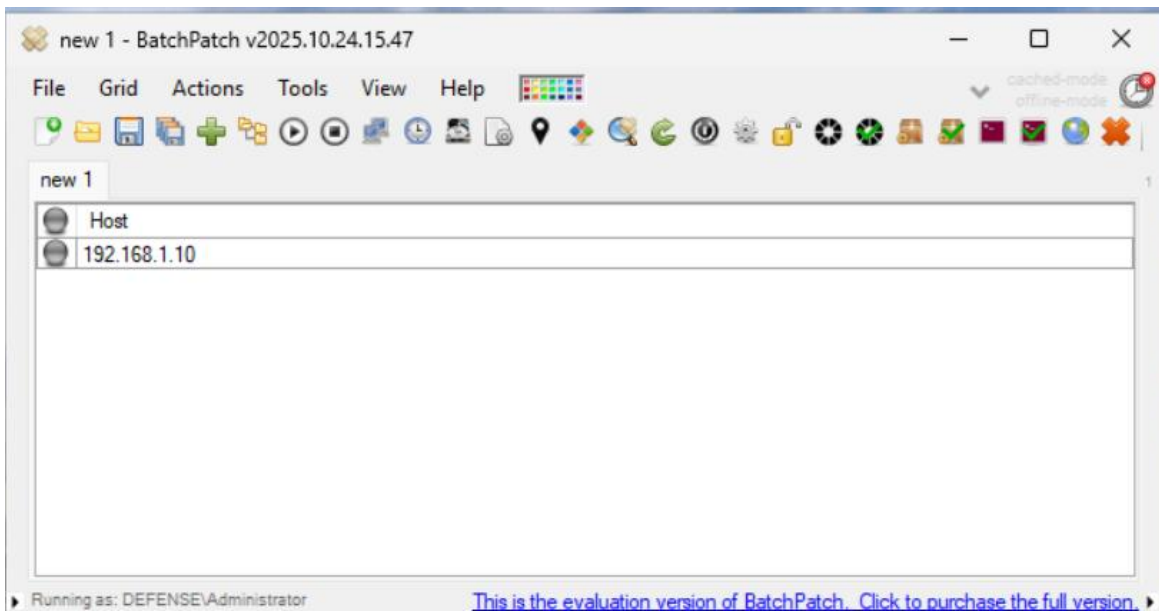
Show: Main interface

Step 3: Add Target (Server)

Steps:

1. Click inside the grid

2. Type IP: 192.168.1.10



E4-03-add-server-ip.png

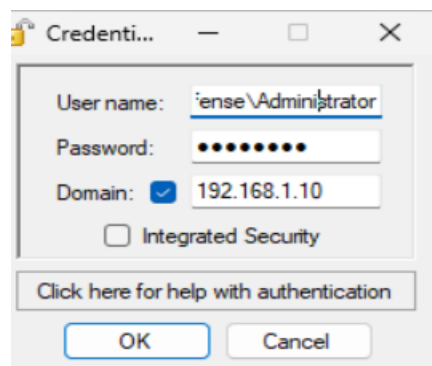
Location: BatchPatch grid

Show: Server IP (192.168.1.10) added to list

Step 4: Configure Domain Credentials

Steps:

1. Right-click on IP → **Specify alternate logon credentials**
2. Enter:
 - Username: defense\Administrator
 - Password: Pa\$\$w0rd



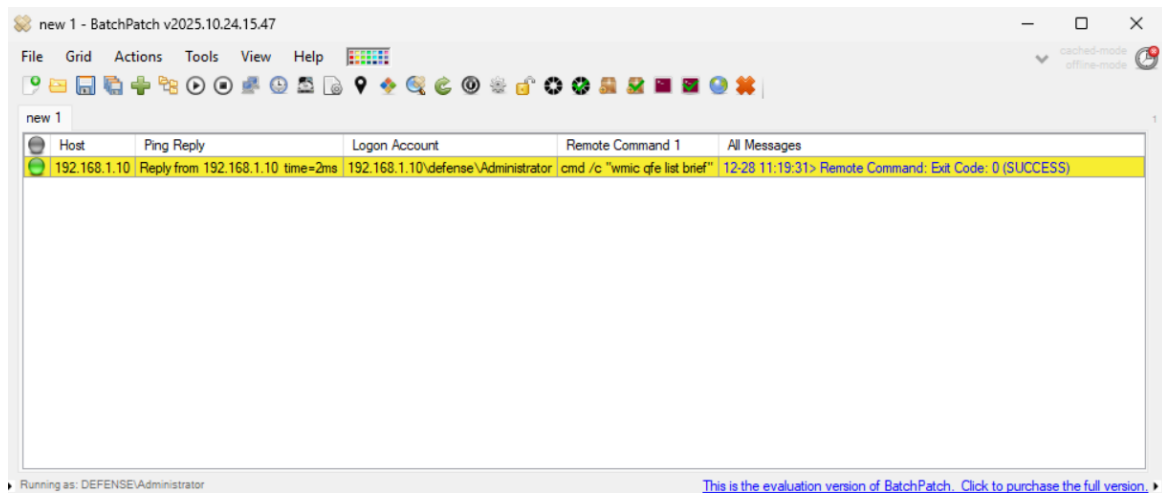
E4-04-domain-credentials.png

Location: Credentials dialog
Show: Domain credentials configured

Step 5: Scan for Windows Updates

Steps:

1. Right-click on IP
2. Select **Windows Updates** → **Check for available updates**



E4-05-scanning.png

Location: BatchPatch
Show: "Installing Remote Agent" or "Scanning" status

Step 6: Review Missing Patches

Result:

- List of Security Updates
- Critical / Important Updates

To view installed patches remotely, you can use the WMIC command:

```
cmd /c "wmic qfe list brief"
```

This command lists all Quick Fix Engineering (QFE) hotfixes installed on the system.



E4-06-missing-patches.png

Location: BatchPatch results

Show: List of missing patches with severity

🔒 **Security Note:** Unpatched systems are one of the primary vectors for endpoint exploitation.

Summary Table

Item	Status
Remote Agent	Installed
Connectivity	Success
Missing Patches	Detected
Admin Credential	Domain Admin

Table 10: BatchPatch Summary

Exercise 5: Patch Manager Plus (ManageEngine)

Target: Windows Client (Management + Agent)

Objective:

- Deploy centralized patch management
 - Identify vulnerabilities from missing patches
 - Gain experience with enterprise patch management tool
-

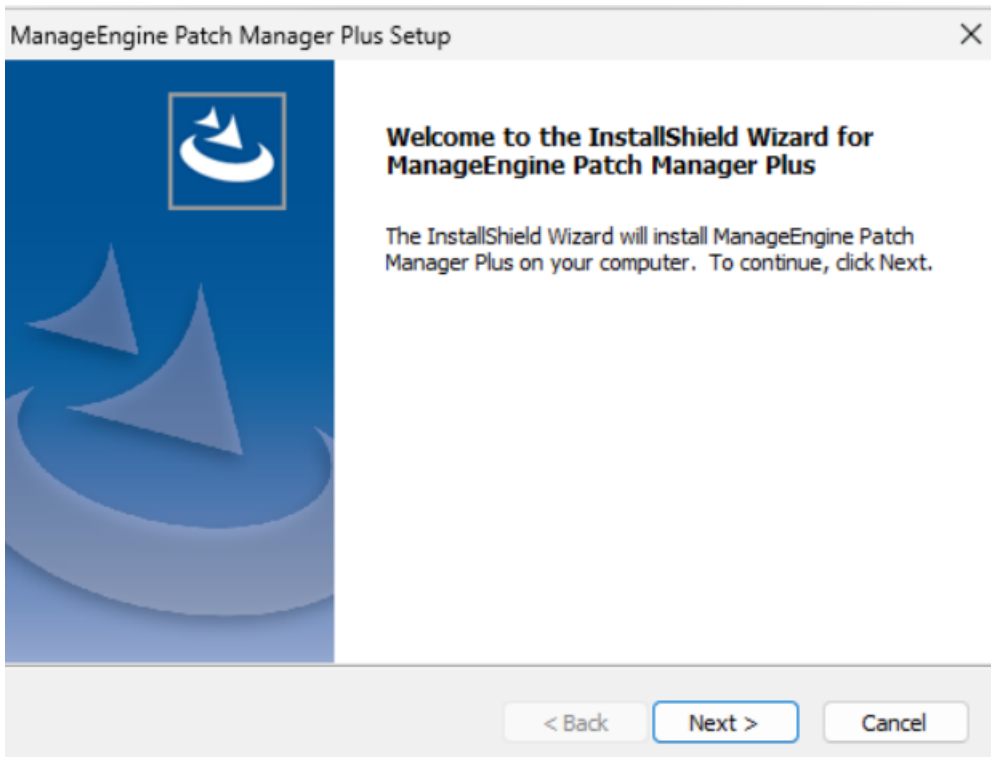
Step 1: Download and Install Patch Manager Plus

Location: Windows Client

Download: [ManageEngine Patch Manager Plus](#) (Trial)

Installation:

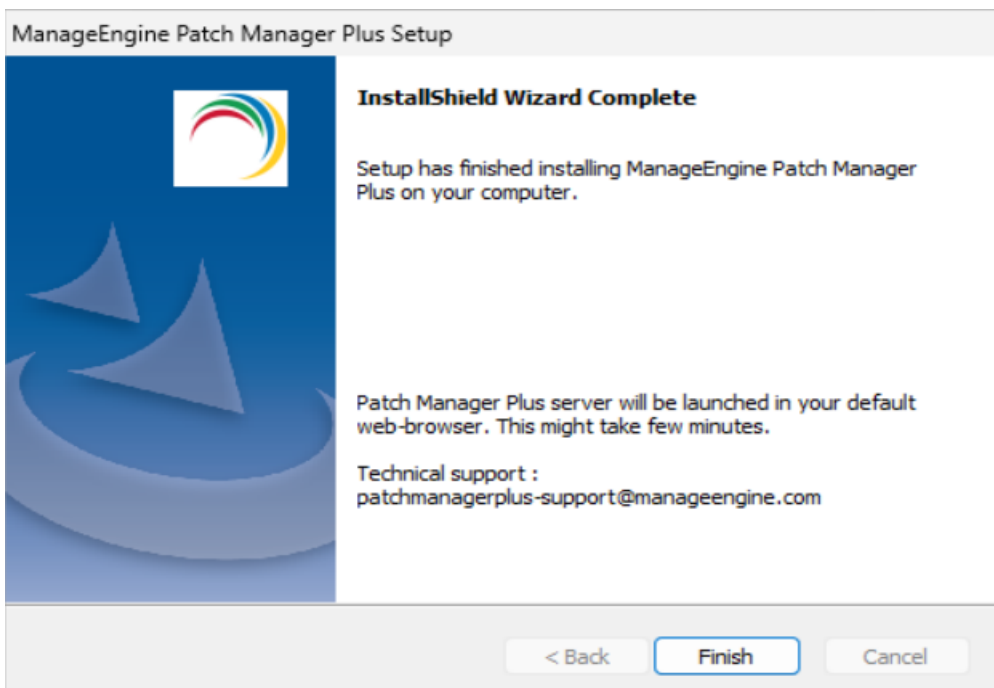
1. Run Installer
2. Select: **On-Premise**
3. Complete wizard



E5-01-onprem-install.png

Location: Installer

Show: On-Premise deployment type selected



E5-02-install-finished.png

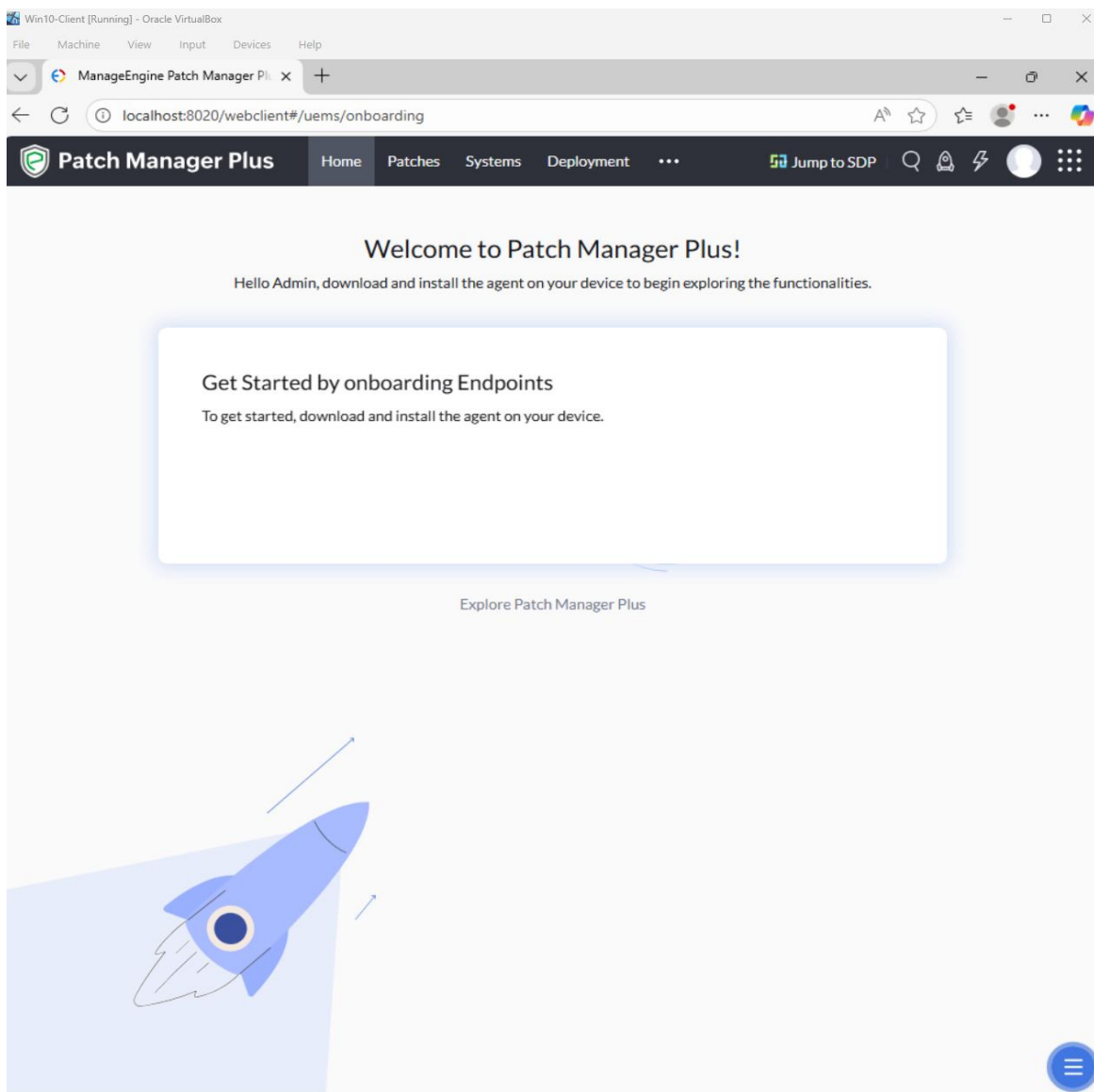
Location: Installer
Show: Installation complete

Step 2: Access Web Console

URL: `http://localhost:8020`

Credentials:

- Username: `admin`
- Password: `admin`



E5-03-dashboard.png

Location: Web Console

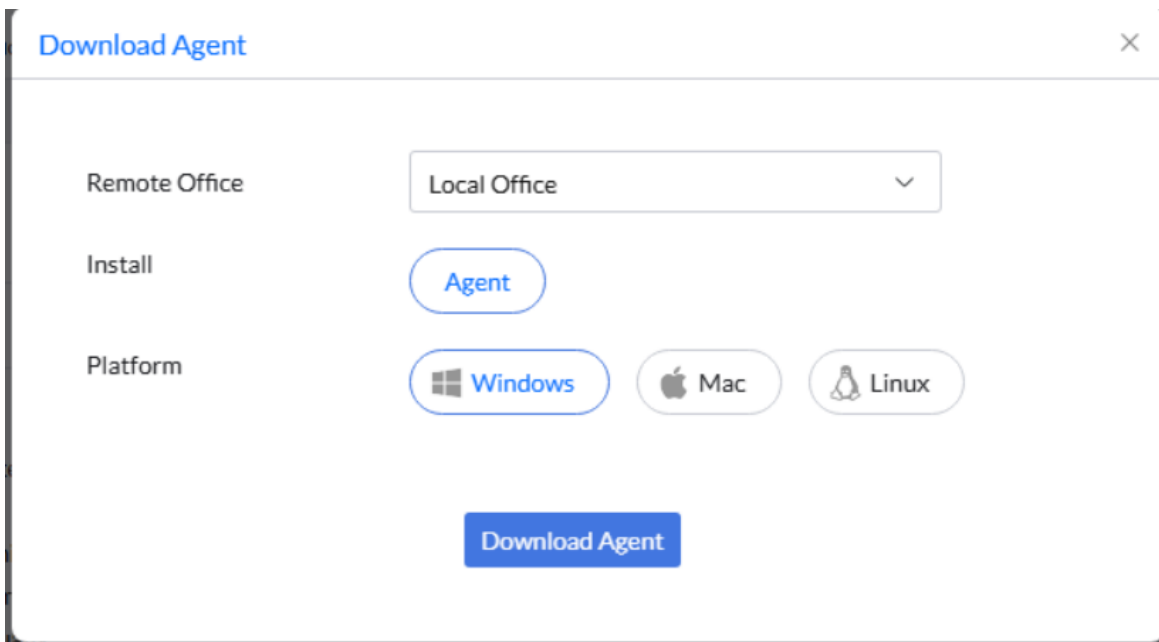
Show: Main dashboard

🔒 **Security Note:** Centralized console allows control of patches across all endpoints.

Step 3: Add Computers (Install Agent)

Steps:

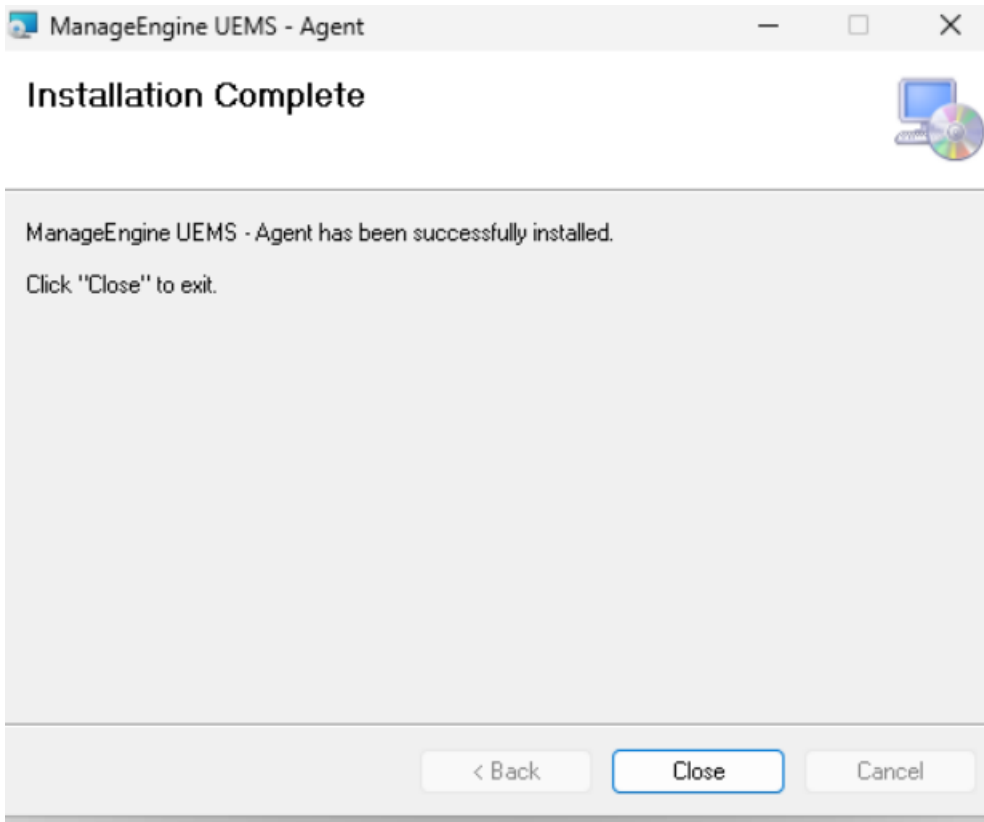
1. Go to **Agent** → **SoM (Scope of Management)** → **Add Computers**
2. Select domain/workgroup, choose the computers to manage, or manually enter names
3. Enable "Install Agent Immediately" if you want instant deployment



E5-04-download-agent.png

Location: Web Console

Show: Add Computers screen under Agent SoM



E5-05-agent-installed.png

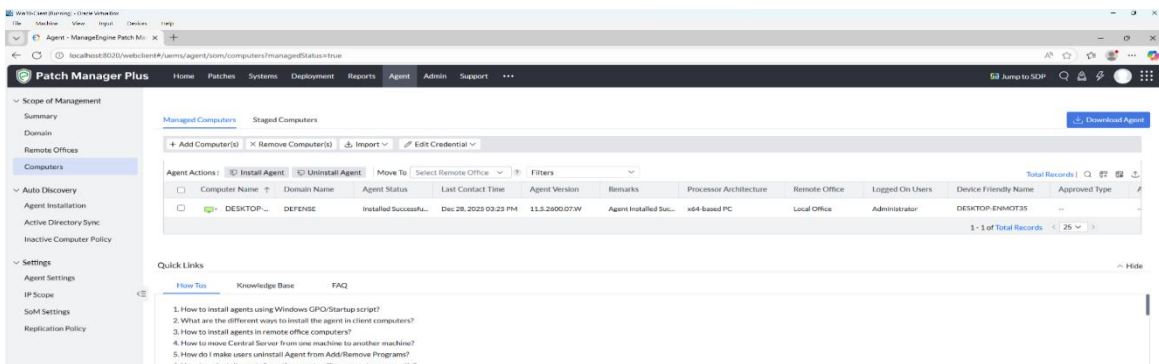
Location: Agent installer

Show: Agent installation complete

Step 4: Verify System Detection

Location: Web Console → **Agent** → **SoM** → **Managed Computers**

Expected Result: ✓ Client appears as Online



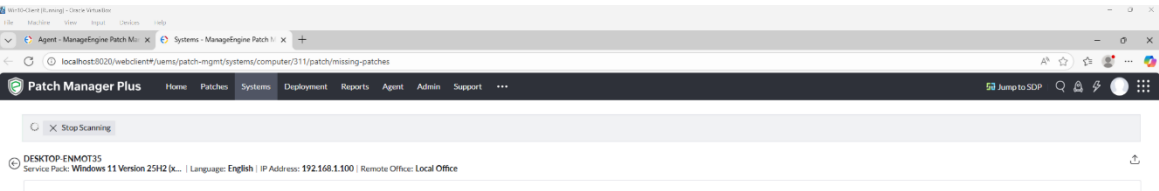
E5-06-client-visible.png

Location: Managed Computers list
Show: Client listed as online

Step 5: Scan for Patches

Steps:

- 1. Select Client
- 2. Click **Scan Systems** → **Scan Now**



E5-07-scan-running.png

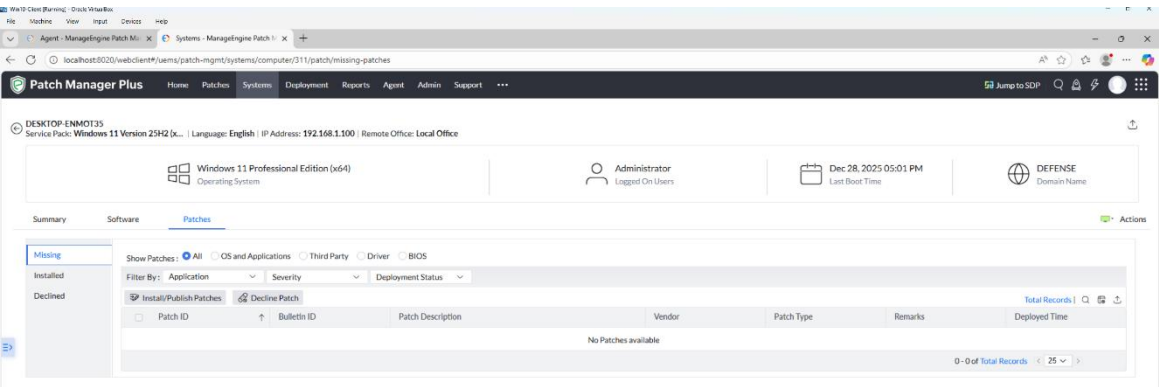
Location: Web Console
Show: Scanning status

Step 6: Analyze Missing Patches

Location: Missing Patches tab

Shows:

- Security Updates
- Severity (Critical / Important)



E5-08-missing-patches.png

Location: Missing Patches view
Show: List of missing patches with severity ratings

🛡️ **Security Analysis:** Patch Manager Plus identified uninstalled security patches that could lead to endpoint exploitation. Centralized patch management significantly reduces vulnerability-based attack risk.

Summary Table

Item	Status
Console	Installed
Agent	Installed
Scan	Successful
Missing Patches	Detected

Table 11: Patch Manager Plus Summary

Exercise 6: Delegation of Control

Target: Windows Server 2019 (Domain Controller)

Objective:

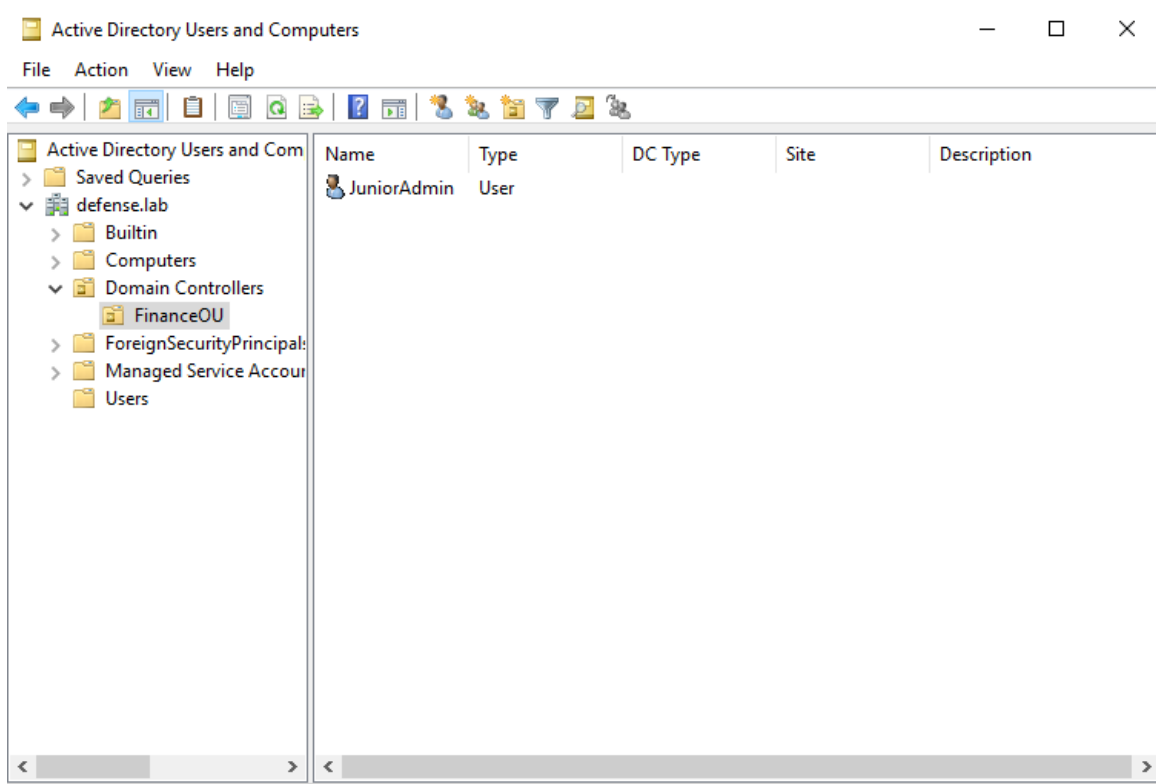
- Implement Principle of Least Privilege
 - Create Junior Admin with limited permissions
 - Practice AD delegation
-

Step 1: Create Junior Admin User

Location: Windows Server 2019

Steps:

1. Open **Server Manager** → **Tools** → **Active Directory Users and Computers**
2. Right-click on OU (e.g., **FinanceOU**) → **New** → **User**
3. Create user:
 - Username: JuniorAdmin
 - Password: Pa\$\$w0rd



E6-01-junioradmin-created.png

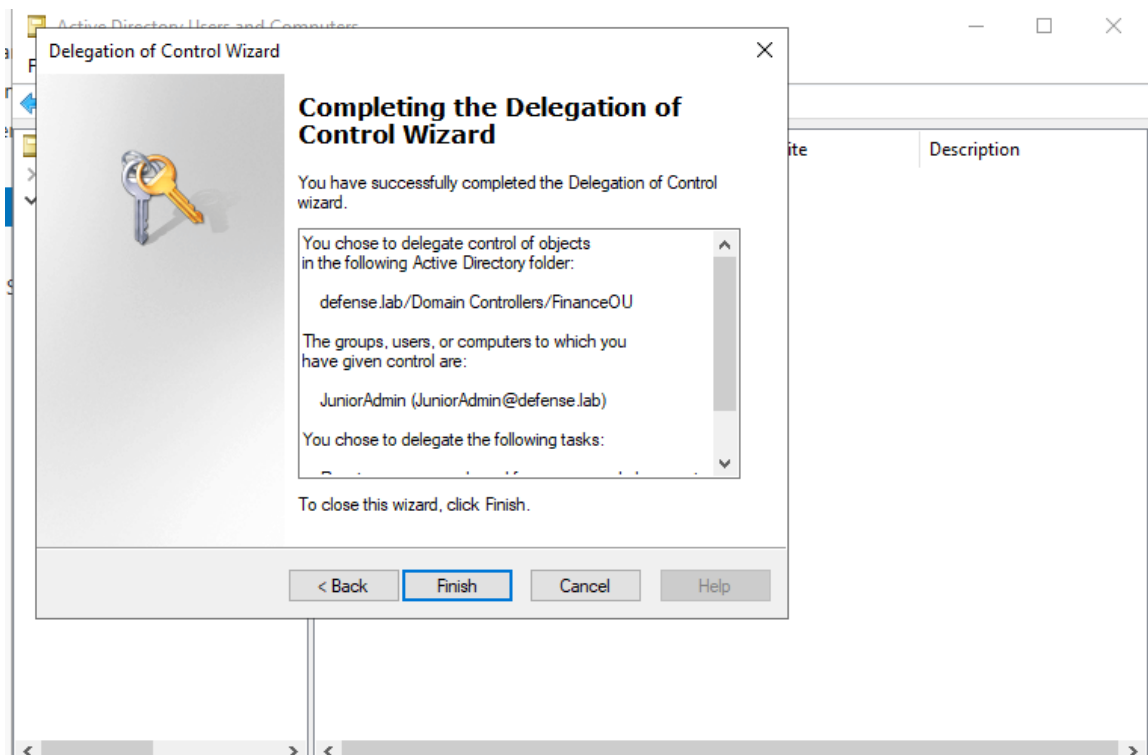
Location: Active Directory Users and Computers

Show: JuniorAdmin user created

Step 2: Delegate Control

Steps:

1. Right-click **FinanceOU** → **Delegate Control**
2. Click **Add** → **JuniorAdmin**
3. Select task: **Reset user passwords and force password change at next logon**
4. Click **Finish**



E6-02-delegate-wizard.png

Location: Delegation of Control Wizard

Show: Task selection screen

Step 3: Test Junior Admin Access

Location: Windows Client (Domain-joined)

Login:

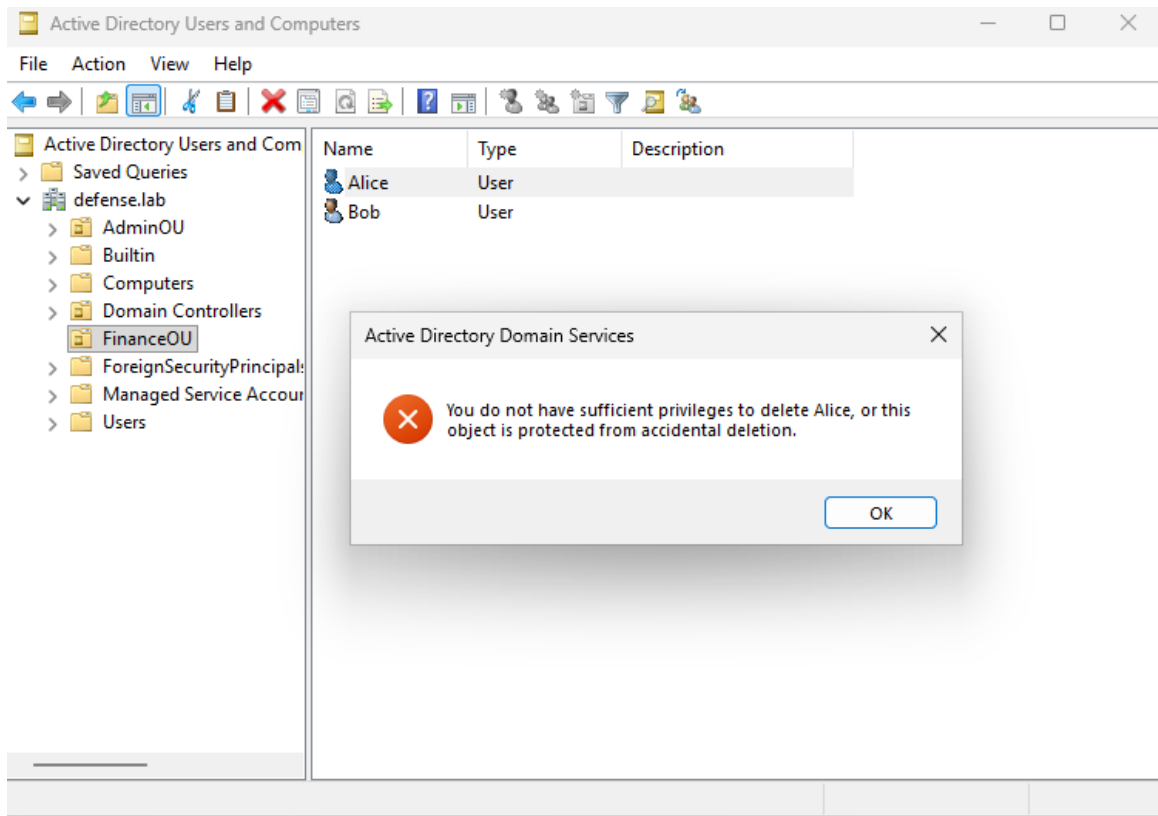
- Username: defense\JuniorAdmin
- Password: Pa\$\$w0rd

Test 1: Reset Password

- Open RSAT → **Active Directory Users and Computers**
- Try to reset a user password in FinanceOU
- **Expected Result:** ✓ Success

Test 2: Delete User

- Try to delete a user in FinanceOU
- **Expected Result:** ✗ Fail (Access Denied)



E6-04-delete-user-fail.png

Location: ADUC

Show: Delete user operation denied

Access Summary Table

User	Task	Result
JuniorAdmin	Reset Password	Success
JuniorAdmin	Delete User	Fail

Table 12: Delegation of Control Access Results

🛡️ **Security Analysis:** Through delegation, JuniorAdmin has only the necessary permissions. Principle of Least Privilege is maintained, reducing risk of abuse.

Exercise 7: LAPS (Local Administrator Password Solution)

Targets: Windows Server 2019 (Management) & Windows Client (Target)

Objective:

- Secure Local Administrator passwords
- Prevent abuse of static admin accounts
- Use GPO for security automation

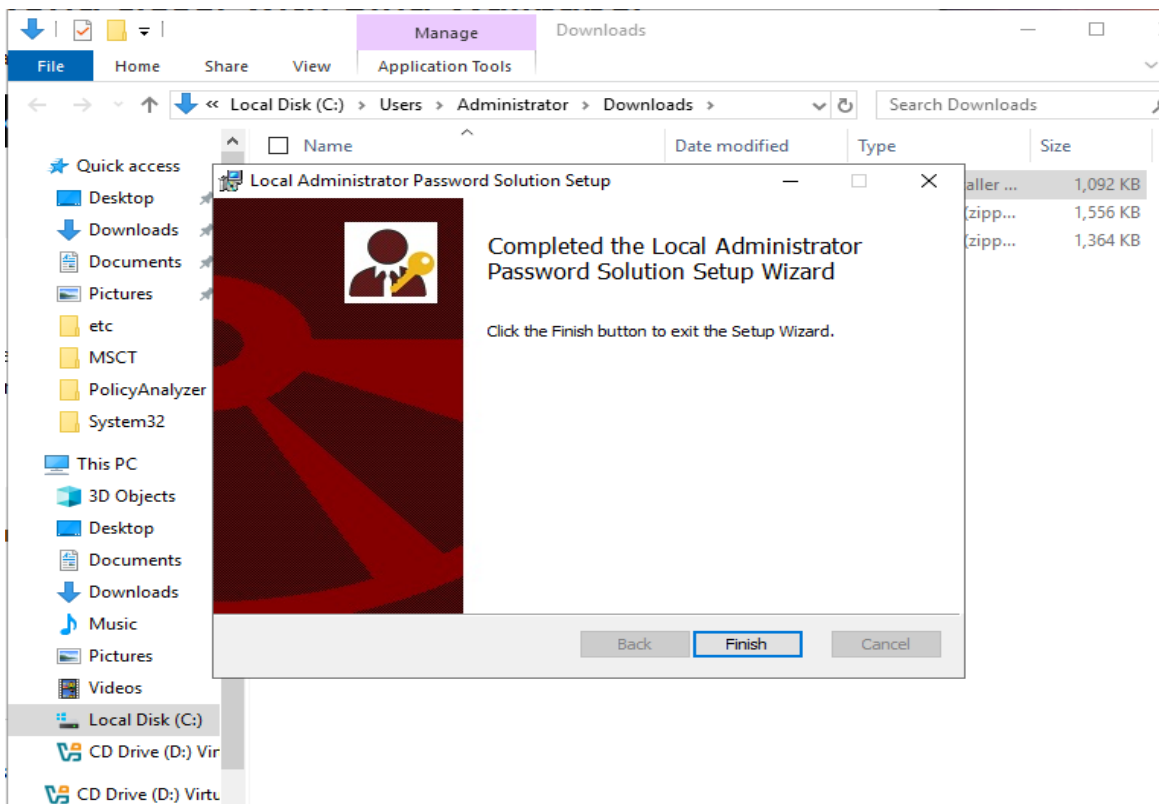
Step 1: Install LAPS (Server)

Location: Windows Server 2019

Download: [LAPS.x64.msi](#)

Installation:

1. Run installer
2. Select features:
 - Management Tools
 - PowerShell Module
 - GPO Templates



E7-01-laps-server-install.png

Location: LAPS Installer

Show: Feature selection (Management Tools, PowerShell, GPO Templates)

I can't Install LAPS in cinet to continue the othe steps.

Exercise 8: Endpoint Central (Application Control)

Target: Windows Client

Objective:

- Manage application execution on endpoints
 - Simulate Application Whitelisting/Blacklisting
 - Enhance endpoint security and prevent malware execution
-

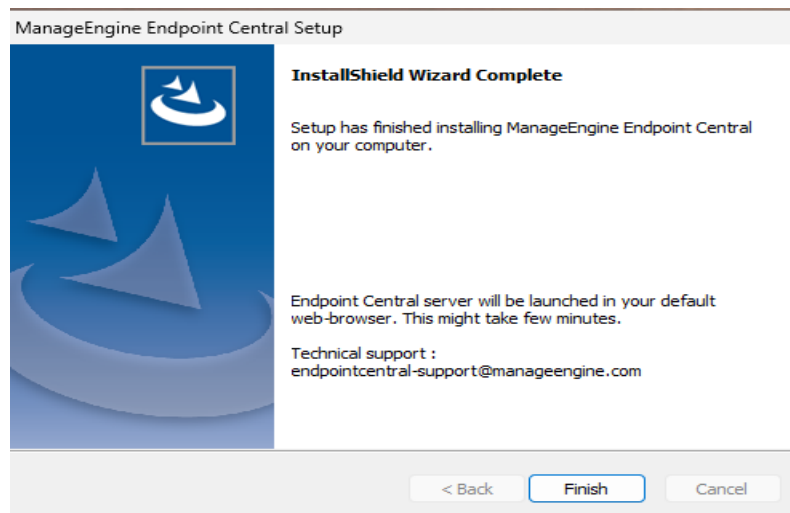
Step 1: Download and Install Endpoint Central

Location: Windows Client (or Server)

Download: [ManageEngine Endpoint Central](#) (Trial)

Installation:

1. Run installer
2. Complete installation wizard



E8-01-install-endpoint-central.png

Location: Installer

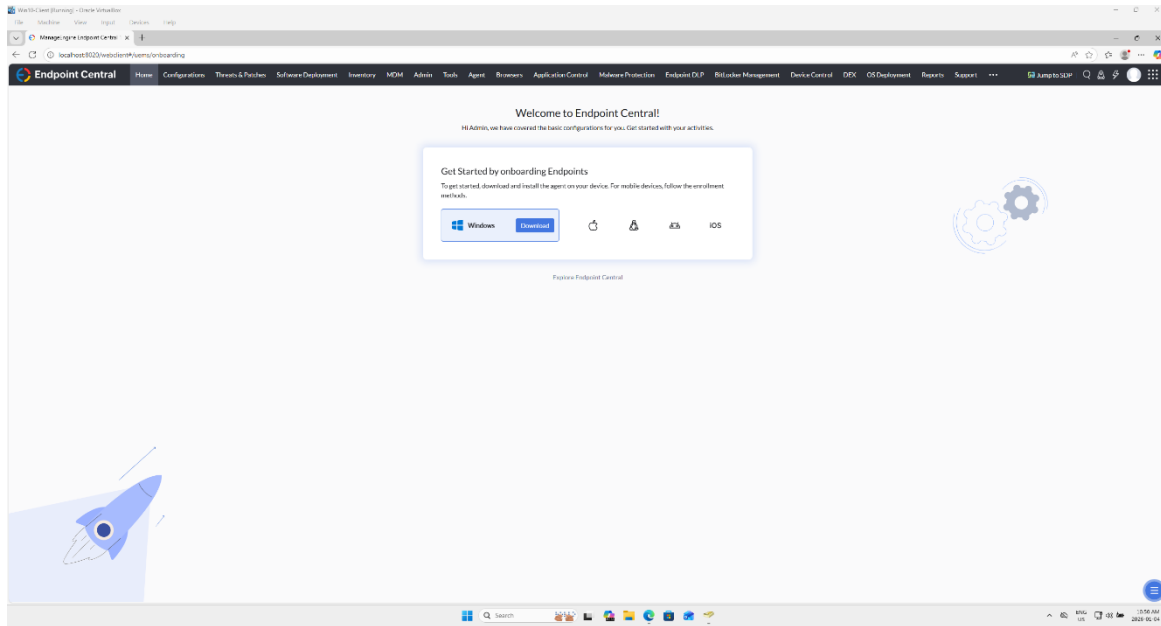
Show: Installation progress or completion

Step 2: Access Web Console

URL: `http://localhost:8020`

Credentials:

- Username: `admin`
- Password: `admin`



E8-02-dashboard.png

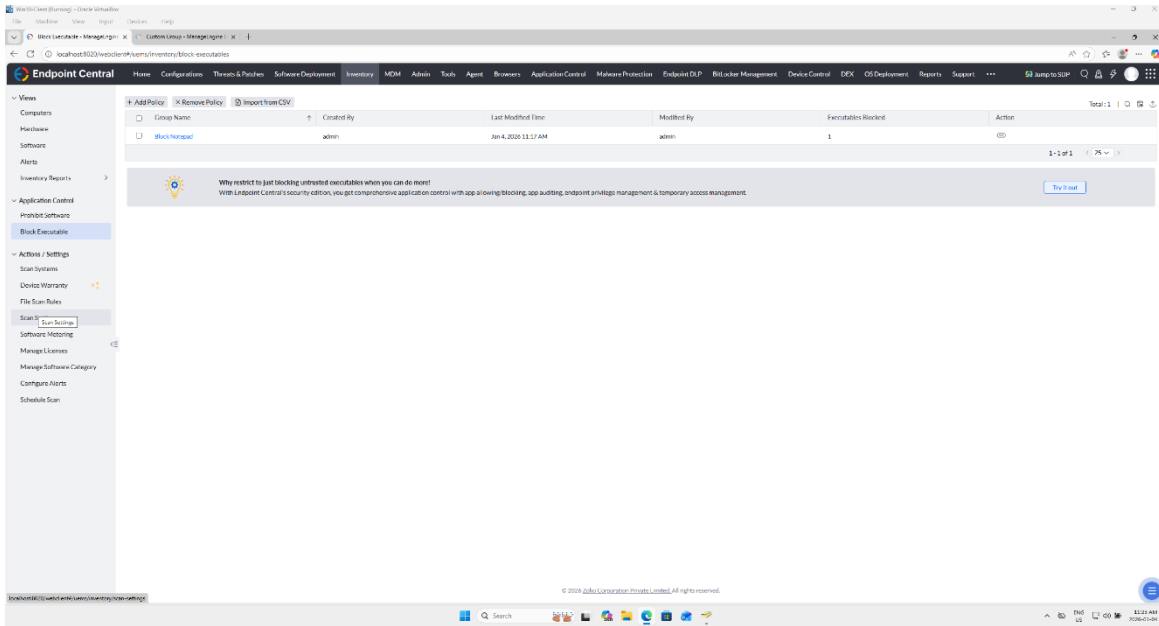
Location: Web Console

Show: Main dashboard

Step 3: Block Software

Steps:

1. Go to **Inventory** → **Application Control**
2. Select **Block Executable**
3. Choose application to block (e.g., `notepad.exe` or `firefox.exe`)
4. Create policy
5. Apply policy to test machine group (e.g., "Local Office")



E8-03-block-policy.png

Location: Application Control

Show: Policy created with blocked application

Step 4: Force Agent Refresh and Test

Steps:

1. Force refresh policy from Web Console
2. Attempt to run blocked application on Client

Expected Result: ✖ Application is prevented from executing

Summary Table

Item	Action	Result
notepad.exe	Blocked	Cannot Run
firefox.exe	Blocked	Cannot Run
Policy Applied	Local Office group	Success

Table 14: Application Control Results

🛡️ **Security Analysis:** Application Control prevents unauthorized or malicious programs from executing on endpoints. This is a key component of Defense in Depth and Application Whitelisting strategies.

Exercise 9: File Integrity Monitoring (OpManager)

Target: Windows Client

Objective:

- Monitor file and folder changes
 - Detect tampering or unauthorized modifications
 - Core component of Endpoint Security & SIEM readiness
-

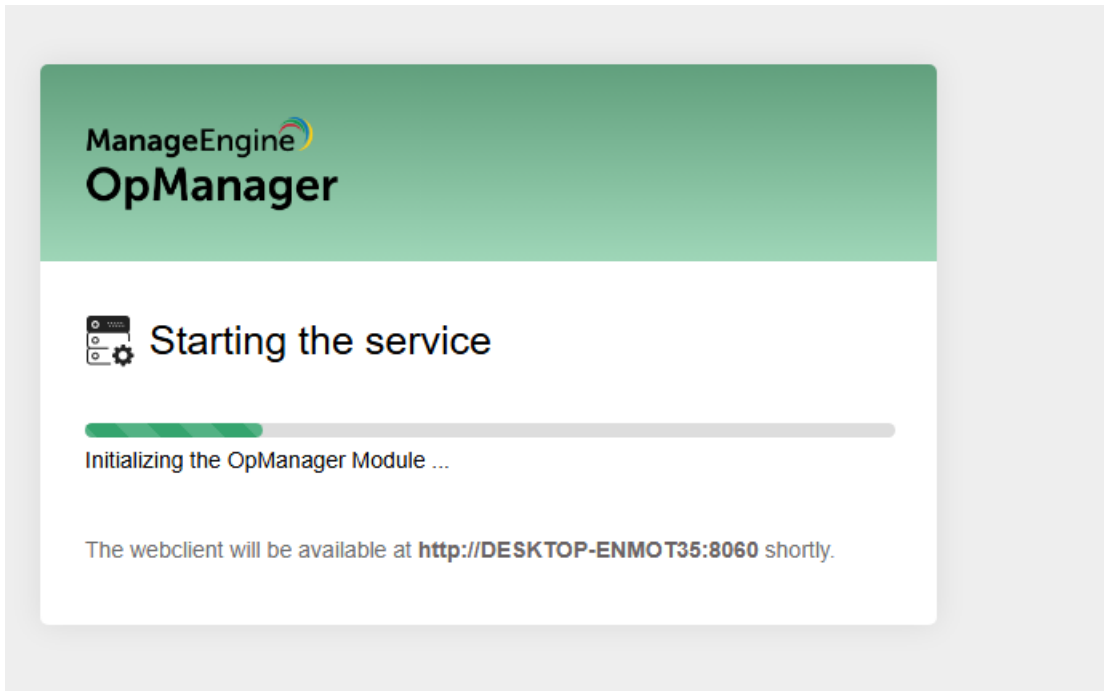
Step 1: Download and Install OpManager

Location: Windows Client

Download: [ManageEngine OpManager](#) (Trial)

Installation:

1. Run installer
2. Complete installation
3. Launch Web Console



E9-01-opmanager-install.png

Location: Installer

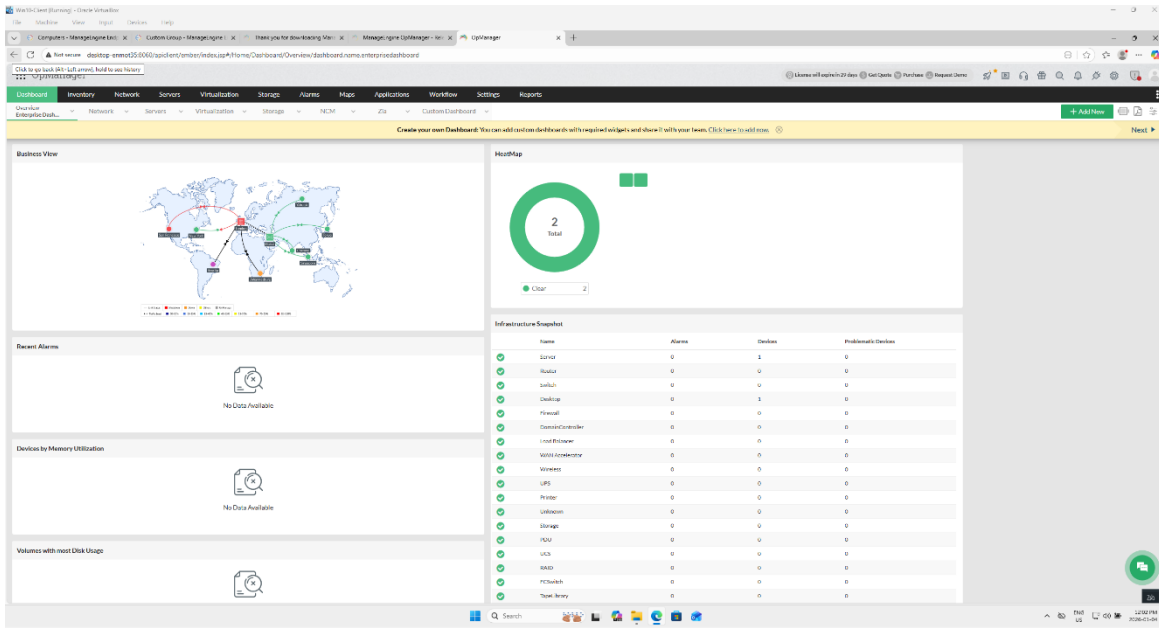
Show: Installation progress or completion

Step 2: Access Web Console

URL: <http://localhost:8060>

Credentials:

- Username: admin
- Password: admin



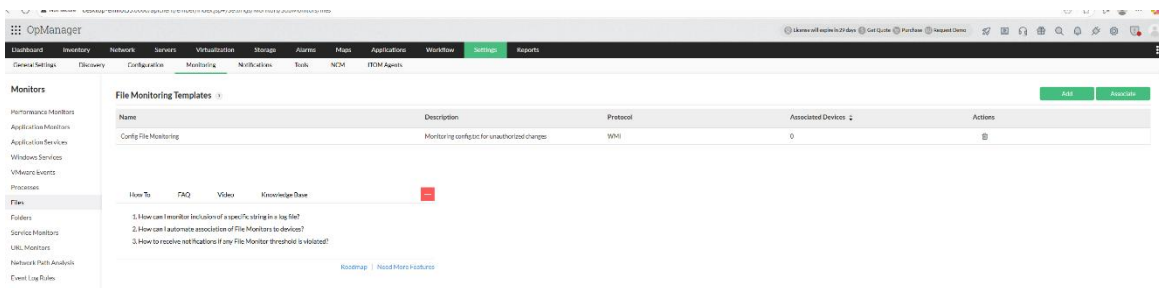
E9-02-dashboard.png

Location: Web Console
Show: Main dashboard

Step 3: Configure File Integrity Monitoring (FIM)

Steps:

1. Go to **Settings** → **Monitoring** → **File Integrity**
2. Add file/folder to monitor
3. Example: Create C:\Critical\config.txt
4. Add to monitoring list
5. Configure alert on change



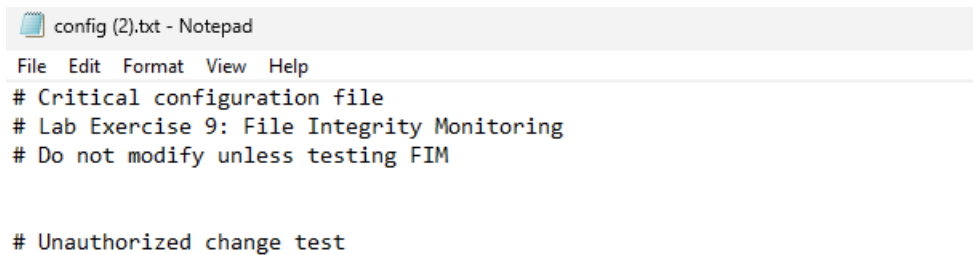
E9-03-fim-add-file.png

Location: File Integrity settings
Show: config.txt added to monitoring

Step 4: Simulate File Modification

Steps:

1. Open C:\Critical\config.txt
2. Change contents (simulate tampering)
3. Save file



```
config (2).txt - Notepad
File Edit Format View Help
# Critical configuration file
# Lab Exercise 9: File Integrity Monitoring
# Do not modify unless testing FIM

# Unauthorized change test
```

E9-04-file-modified.png

Location: Notepad or file comparison
Show: File before/after modification

Step 5: Detect Change in OpManager

Steps:

1. In Web Console, click **Scan Now**
2. Check **Alarms** section

Expected Result:

- Alert generated
 - Checksum changed
 - Access and modification report displayed
-

Summary Table

File	Action	Result
config.txt	Modified	Alert Generated
Checksum	Changed	Detected

Table 15: File Integrity Monitoring Results

🛡️ **Security Analysis:** FIM enables detection of unauthorized changes and is a foundational component of Endpoint Detection & Response (EDR). This technique helps prevent and detect malware activity and unauthorized modifications.
